

Deploying Installer Provisioned Infrastructure (IPI) of OpenShift on Bare Metal - 4.6

Deployment Integration Team

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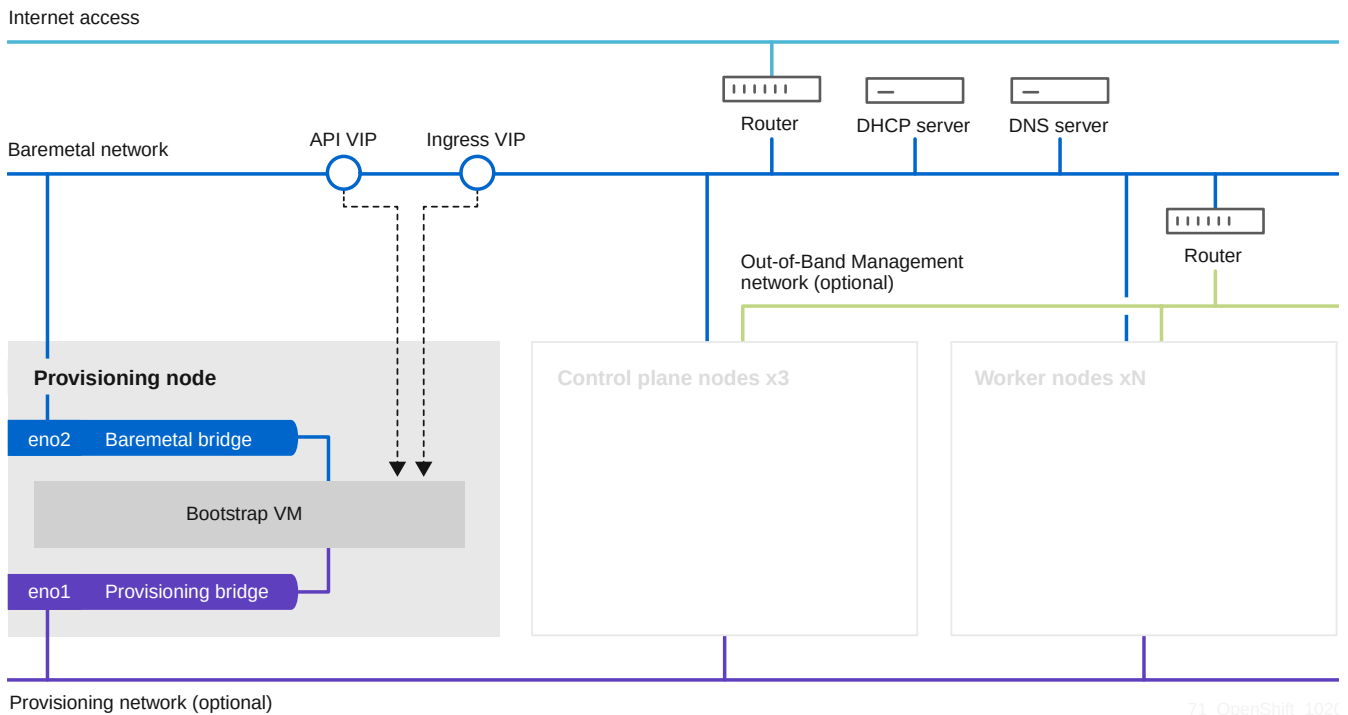


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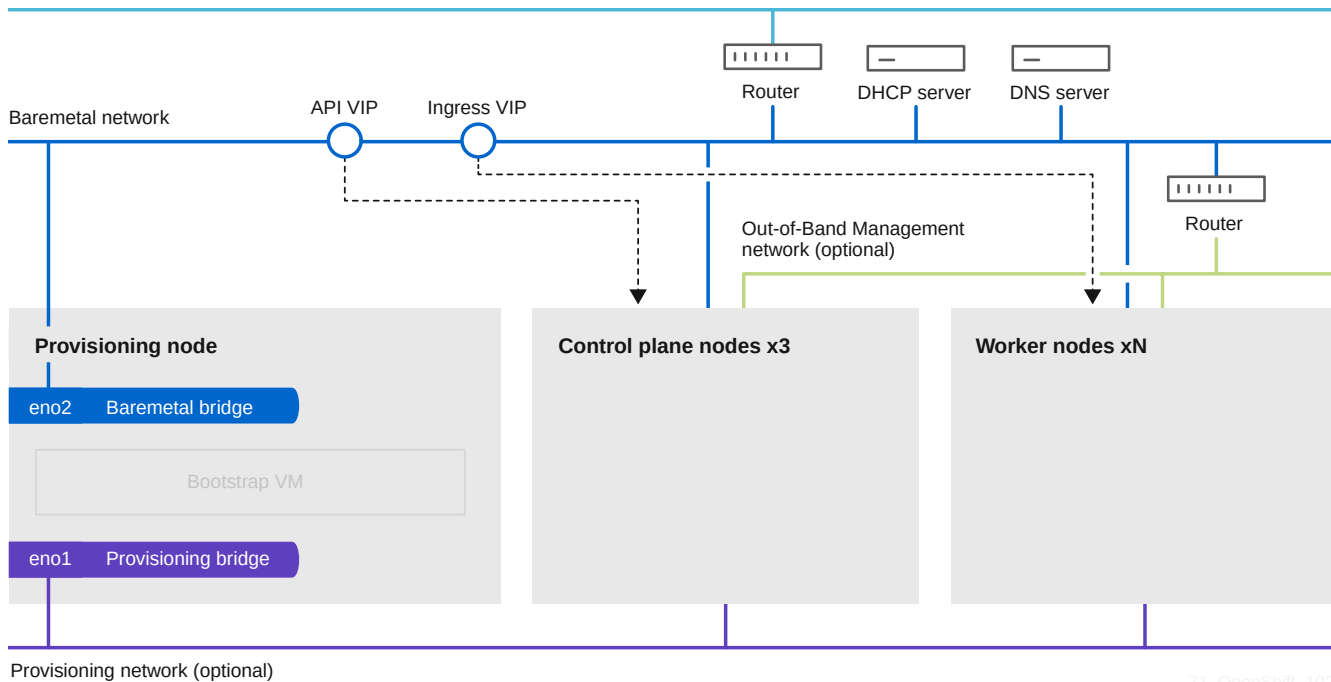
Chapter 1. Overview

Installer-provisioned installation provides support for installing OpenShift Container Platform on bare metal nodes. This guide provides a methodology to achieving a successful installation.

During installer-provisioned installation on bare metal, the installer on the bare metal node labeled as **provisioner** creates a bootstrap virtual machine (VM). The role of the bootstrap VM is to assist in the process of deploying an OpenShift Container Platform cluster. The bootstrap VM connects to the **baremetal** network and to the **provisioning** network, if present, via the network bridges.



When the installation of OpenShift control plane nodes is complete and fully operational, the installer destroys the bootstrap VM automatically and moves the virtual IP addresses (VIPs) to the appropriate nodes. The API VIP moves to the control plane nodes and the Ingress VIP moves to the worker nodes.



Chapter 2. Prerequisites

Installer-provisioned installation of OpenShift Container Platform requires:

1. One provisioner node with Red Hat Enterprise Linux (RHEL) 8.x installed.
2. Three control plane nodes.
3. Baseboard Management Controller (BMC) access to each node.
4. At least one network:
 - a. One **required** routable network
 - b. One **optional** network for provisioning nodes; and,
 - c. One **optional** management network.

Before starting an installer-provisioned installation of OpenShift Container Platform, ensure the hardware environment meets the following requirements.

2.1. Node requirements

Installer-provisioned installation involves a number of hardware node requirements:

- **CPU architecture:** All nodes must use `x86_64` CPU architecture.
- **Similar nodes:** Red Hat recommends nodes have an identical configuration per role. That is, Red Hat recommends nodes be the same brand and model with the same CPU, memory and storage configuration.
- **Baseboard Management Controller:** The `provisioner` node must be able to access the baseboard management controller (BMC) of each OpenShift Container Platform cluster node. You may use IPMI, Redfish, or a proprietary protocol.
- **Latest generation:** Nodes must be of the most recent generation. Installer-provisioned installation relies on BMC protocols, which must be compatible across nodes. Additionally, RHEL 8 ships with the most recent drivers for RAID controllers. Ensure that the nodes are recent enough to support RHEL 8 for the `provisioner` node and RHCOS 8 for the control plane and worker nodes.
- **Registry node:** (Optional) If setting up a disconnected mirrored registry, it is recommended the registry reside in its own node.
- **Provisioner node:** Installer-provisioned installation requires one `provisioner` node.
- **Control plane:** Installer-provisioned installation requires three control plane nodes for high availability.
- **Worker nodes:** While not required, a typical production cluster has one or more worker nodes. Smaller clusters are more resource efficient for administrators and developers during development, production, and testing.
- **Network interfaces:** Each node must have at least one 10GB network interface for the routable `baremetal` network. Each node must have one 10GB network interface for a `provisioning` network **when using the provisioning network** for deployment. Using the `provisioning`

network is the default configuration. Network interface names must follow the same naming convention across all nodes. For example, the first NIC name on a node, such as `eth0` or `eno1`, must be the same name on all of the other nodes. The same principle applies to the remaining NICs on each node.

- **Unified Extensible Firmware Interface (UEFI):** Installer-provisioned installation requires UEFI boot on all OpenShift Container Platform nodes when using IPv6 addressing on the `provisioning` network. In addition, UEFI Device PXE Settings must be set to use the IPv6 protocol on the `provisioning` network NIC, but omitting the `provisioning` network removes this requirement.

2.2. Firmware requirements for installing with virtual media

The installer for installer-provisioned OpenShift Container Platform clusters validates the hardware and firmware compatibility with Redfish virtual media. The following table lists supported firmware for installer-provisioned OpenShift Container Platform clusters deployed with Redfish virtual media.

Table 1. Firmware compatibility for Redfish virtual media

Hardware	Model	Management	Firmware Versions
HP	10th Generation	iLO5	N/A
	9th Generation	iLO4	N/A
Dell	14th Generation	iDRAC 9	v4.20.20.20 - 04.40.00.00
	13th Generation	iDRAC 8	v2.75.75.75+

Refer to the hardware documentation for the nodes or contact the hardware vendor for information on updating the firmware.

There are no known firmware limitations for HP servers.



For Dell servers, ensure the OpenShift Container Platform cluster nodes have AutoAttach Enabled through the iDRAC console. The menu path is: **Configuration** → **Virtual Media** → **Attach Mode** → **AutoAttach**. With iDRAC 9 firmware version `04.40.00.00`, the Virtual Console plug-in defaults to `eHTML5`, which causes problems with the **InsertVirtualMedia** workflow. Set the plug-in to `HTML5` to avoid this issue. The menu path is: **Configuration** → **Virtual console** → **Plug-in Type** → **HTML5**.



The installer will not initiate installation on a node if the node firmware is below the foregoing versions when installing with virtual media.

2.3. Network requirements

Installer-provisioned installation of OpenShift Container Platform involves several network requirements by default. First, installer-provisioned installation involves a non-routable

provisioning network for provisioning the operating system on each bare metal node and a routable **baremetal** network. Since installer-provisioned installation deploys **ironic-dnsmasq**, the networks should have no other DHCP servers running on the same broadcast domain. Network administrators must reserve IP addresses for each node in the OpenShift Container Platform cluster.

Network Time Protocol (NTP)

Each OpenShift Container Platform node in the cluster must have access to an NTP server. OpenShift Container Platform nodes use NTP to synchronize their clocks. For example, cluster nodes use SSL certificates that require validation, which might fail if the date and time between the nodes are not in sync.



Define a consistent clock date and time format in each cluster node's BIOS settings, or installation might fail.

Configuring NICs

OpenShift Container Platform deploys with two networks:

- **provisioning**: The **provisioning** network is an **optional** non-routable network used for provisioning the underlying operating system on each node that is a part of the OpenShift Container Platform cluster. The network interface for the **provisioning** network on each cluster node must have the BIOS or UEFI configured to PXE boot. In OpenShift Container Platform 4.3, when deploying using the **provisioning** network, the first NIC on each node, such as **eth0** or **eno1**, **must** interface with the **provisioning** network. In OpenShift Container Platform 4.4 and later releases, you can specify the provisioning network NIC with the **provisioningNetworkInterface** configuration setting.
- **baremetal**: The **baremetal** network is a routable network. In OpenShift Container Platform 4.3, when deploying using the **provisioning** network, the second NIC on each node, such as **eth1** or **eno2**, **must** interface with the **baremetal** network. In OpenShift Container Platform 4.4 and later releases, you can use any NIC order to interface with the **baremetal** network, provided it is the same NIC order across worker and control plane nodes and not the NIC specified in the **provisioningNetworkInterface** configuration setting for the **provisioning** network.



Use a compatible approach such that cluster nodes use the same NIC ordering on all cluster nodes. NICs must have heterogeneous hardware with the same NIC naming convention such as **eth0** or **eno1**.



When using a VLAN, each NIC must be on a separate VLAN corresponding to the appropriate network.

Configuring the DNS server

Clients access the OpenShift Container Platform cluster nodes over the **baremetal** network. A network administrator must configure a subdomain or subzone where the canonical name extension is the cluster name.

```
<cluster-name>.<domain-name>
```

For example:

```
test-cluster.example.com
```

For assistance in configuring the DNS server, check [Appendix](#) section for:

- [Creating DNS Records with Bind \(Option 1\)](#)
- [Creating DNS Records with dnsmasq \(Option 2\)](#)

Reserving IP addresses for nodes with the DHCP server

For the **baremetal** network, a network administrator must reserve a number of IP addresses, including:

1. Two virtual IP addresses.
 - One IP address for the API endpoint
 - One IP address for the wildcard ingress endpoint
2. One IP address for the provisioner node.
3. One IP address for each control plane (master) node.
4. One IP address for each worker node, if applicable.

The following table provides an exemplary embodiment of fully qualified domain names. The API and Nameserver addresses begin with canonical name extensions. The host names of the control plane and worker nodes are exemplary, so you can use any host naming convention you prefer.

Usage	Host Name	IP
API	<code>api.<cluster-name>.<domain></code>	<code><ip></code>
Ingress LB (apps)	<code>*.apps.<cluster-name>.<domain></code>	<code><ip></code>
Provisioner node	<code>provisioner.<cluster-name>.<domain></code>	<code><ip></code>
Master-0	<code>openshift-master-0.<cluster-name>.<domain></code>	<code><ip></code>
Master-1	<code>openshift-master-1.<cluster-name>.<domain></code>	<code><ip></code>
Master-2	<code>openshift-master-2.<cluster-name>.<domain></code>	<code><ip></code>
Worker-0	<code>openshift-worker-0.<cluster-name>.<domain></code>	<code><ip></code>
Worker-1	<code>openshift-worker-1.<cluster-name>.<domain></code>	<code><ip></code>
Worker-n	<code>openshift-worker-n.<cluster-name>.<domain></code>	<code><ip></code>

For assistance in configuring the DHCP server, check [Appendix](#) section for:

- [Creating DHCP reservations with dhcpd \(Option 1\)](#)
- [Creating DHCP reservations with dnsmasq \(Option 2\)](#)

Additional requirements with no provisioning network

All installer-provisioned installations require a **baremetal** network. The **baremetal** network is a

routable network used for external network access to the outside world. In addition to the IP address supplied to the OpenShift Container Platform cluster node, installations without a **provisioning** network require the following:

- Setting an available IP address from the **baremetal** network to the **bootstrapProvisioningIP** configuration setting within the **install-config.yaml** configuration file.
- Setting an available IP address from the **baremetal** network to the **provisioningHostIP** configuration setting within the **install-config.yaml** configuration file.
- Deploying the OpenShift Container Platform cluster using RedFish Virtual Media/iDRAC Virtual Media.



Configuring additional IP addresses for **bootstrapProvisioningIP** and **provisioningHostIP** is not required when using a **provisioning** network.

IPv6 considerations

SLAAC Addressing

If you do not plan to use SLAAC ^[1] addresses on your OpenShift Container Platform node, then it should be disabled for **baremetal** networks, that means that if your network equipment is configured to send SLAAC addresses when replying to Route Advertisements that behavior should be changed, so it only sends the route and not the SLAAC address.

Install **ndptool** on your system in order to check what your RAs look like:

```
# Turn down/up baremetal iface on a master Node
$ sudo nmcli con down "Wired connection 5" && sudo nmcli con up "Wired connection 5"
Connection 'Wired connection 5' successfully deactivated (D-Bus active path:
/org/freedesktop/NetworkManager/ActiveConnection/1983)
Connection successfully activated (D-Bus active path:
/org/freedesktop/NetworkManager/ActiveConnection/2044)

# ndptool monitor on Helper node
$ sudo ndptool monitor -t ra
NDP payload len 80, from addr: fe80::c0a4:6464:bc3:d657, iface: baremetal.153
  Type: RA
  Hop limit: 64
  Managed address configuration: yes
  Other configuration: no
  Default router preference: medium
  Router lifetime: 0s
  Reachable time: unspecified
  Retransmit time: unspecified
  Source linkaddr: 1c:40:24:1b:0c:34
  Prefix: 2620:52:0:1303::/64, valid_time: 86400s, preferred_time: 14400s, on_link:
yes, autonomous_addr_conf: no, router_addr: no
  Route: ::/0, lifetime: 0s, preference: low
```

The `ndptool monitor` should report `Managed address configuration: yes`.

Network Ranges and Configurations

Different `baremetal` and `provisioning` networks are required for each environment; each environment will have a different IPv6 range for each one of those networks.

In our configuration we used subinterfaces attached to two different physical interfaces, VLAN tagging was done at O.S. level (this required switch ports configured with `trunk` mode).

Our different IPv6 networks were all routable but usually, the only routable networks are the `baremetal` ones.

Keep in mind that `provisioning` networks cannot be in the same broadcast domain, since services such as DHCP are running.



Route Advertisement

Route Advertisement must be enabled for both networks `baremetal` and `provisioning`.

Route Advertisements

As mentioned previously, both the `baremetal` and the `provisioning` networks must have Route Advertisement enabled. For the `baremetal` network, the `radvd` daemon was used, while the `provisioning` network has RA enabled in the Metal³ dnsmasq, so no configuration is needed.

2.4. Configuring nodes

Configuring nodes when using the `provisioning` network

Each node in the cluster requires the following configuration for proper installation.



A mismatch between nodes will cause an installation failure.

While the cluster nodes can contain more than two NICs, the installation process only focuses on the first two NICs:

NIC	Network	VLAN
NIC1	<code>provisioning</code>	<provisioning-vlan>
NIC2	<code>baremetal</code>	<baremetal-vlan>

NIC1 is a non-routable network (`provisioning`) that is only used for the installation of the OpenShift Container Platform cluster.

The Red Hat Enterprise Linux (RHEL) 8.x installation process on the provisioner node might vary. To install Red Hat Enterprise Linux (RHEL) 8.x using a local Satellite server or a PXE server, PXE-enable NIC2.

PXE	Boot order
-----	------------

NIC1 PXE-enabled provisioning network	1
NIC2 baremetal network. PXE-enabled is optional.	2



Ensure PXE is disabled on all other NICs.

Configure the control plane and worker nodes as follows:

PXE	Boot order
NIC1 PXE-enabled (provisioning network)	1

*Configuring nodes without the **provisioning** network*

The installation process requires one NIC:

NIC	Network	VLAN
NICx	baremetal	<baremetal-vlan>

NICx is a routable network (**baremetal**) that is used for the installation of the OpenShift Container Platform cluster, and routable to the internet.

2.5. Out-of-band management

Nodes will typically have an additional NIC used by the Baseboard Management Controllers (BMCs). These BMCs must be accessible from the **provisioner** node.

Each node must be accessible via out-of-band management. When using an out-of-band management network, the **provisioner** node requires access to the out-of-band management network for a successful OpenShift Container Platform 4 installation.

The out-of-band management setup is out of scope for this document. We recommend setting up a separate management network for out-of-band management. However, using the **provisioning** network or the **baremetal** network are valid options.

2.6. Required data for installation

Prior to the installation of the OpenShift Container Platform cluster, gather the following information from all cluster nodes:

- Out-of-band management IP
 - Examples
 - Dell (iDRAC) IP
 - HP (iLO) IP

*When using the **provisioning** network*

- NIC1 (**provisioning**) MAC address

- NIC2 (**baremetal**) MAC address

When omitting the **provisioning** network

- NICx (**baremetal**) MAC address

2.7. Validation checklist for nodes

When using the **provisioning** network

- ☐ NIC1 VLAN is configured for the **provisioning** network.
- ☐ NIC2 VLAN is configured for the **baremetal** network.
- ☐ NIC1 is PXE-enabled on the provisioner, Control Plane (master), and worker nodes.
- ☐ PXE has been disabled on all other NICs.
- ☐ Control plane and worker nodes are configured.
- ☐ All nodes accessible via out-of-band management.
- ☐ A separate management network has been created. (optional)
- ☐ Required data for installation.

When omitting the **provisioning** network

- ☐ NICx VLAN is configured for the **baremetal** network.
- ☐ Control plane and worker nodes are configured.
- ☐ All nodes accessible via out-of-band management.
- ☐ A separate management network has been created. (optional)
- ☐ Required data for installation.

Summary

After an environment has been prepared according to the documented prerequisites, the installation process is the same as other installer-provisioned platforms.

[1] Stateless Address AutoConfiguration

Chapter 3. Setting up the environment for an OpenShift installation

3.1. Installing RHEL on the provisioner node

With the networking configuration complete, the next step is to install RHEL 8.X on the provisioner node. The installer uses the provisioner node as the orchestrator while installing the OpenShift Container Platform cluster. For the purposes of this document, installing RHEL on the provisioner node is out of scope. However, options include but are not limited to using a RHEL Satellite server, PXE, or installation media.

3.2. Preparing the provisioner node for OpenShift Container Platform installation

Perform the following steps to prepare the environment.

Procedure

1. Log in to the provisioner node via `ssh`.
2. Create a non-root user (`kni`) and provide that user with `sudo` privileges.

```
[root@provisioner ~]# useradd kni
[root@provisioner ~]# passwd kni
[root@provisioner ~]# echo "kni ALL=(root) NOPASSWD:ALL" | tee -a
/etc/sudoers.d/kni
[root@provisioner ~]# chmod 0440 /etc/sudoers.d/kni
```

3. Create an `ssh` key for the new user.

```
[root@provisioner ~]# su - kni -c "ssh-keygen -t rsa -f /home/kni/.ssh/id_rsa -N
''"
```

4. Log in as the new user on the provisioner node.

```
[root@provisioner ~]# su - kni
[kni@provisioner ~]$
```

5. Use Red Hat Subscription Manager to register the provisioner node.

```
[kni@provisioner ~]$ sudo subscription-manager register --username=<user>
--password=<pass> --auto-attach
[kni@provisioner ~]$ sudo subscription-manager repos --enable=rhel-8-for-x86_64-
```

```
appstream-rpms --enable=rhel-8-for-x86_64-baseos-rpms
```



For more information about Red Hat Subscription Manager, see [Using and Configuring Red Hat Subscription Manager](#).

6. Install the following packages.

```
[kni@provisioner ~]$ sudo dnf install -y libvirt qemu-kvm mkisofs python3-devel jq ipmitool
```

7. Modify the user to add the `libvirt` group to the newly created user.

```
[kni@provisioner ~]$ sudo usermod --append --groups libvirt <user>
```

8. Restart `firewalld` and enable the `http` service.

```
[kni@provisioner ~]$ sudo systemctl start firewalld
[kni@provisioner ~]$ sudo firewall-cmd --zone=public --add-service=http --permanent
[kni@provisioner ~]$ sudo firewall-cmd --add-port=5000/tcp --zone=libvirt --permanent
[kni@provisioner ~]$ sudo firewall-cmd --add-port=5000/tcp --zone=public --permanent
[kni@provisioner ~]$ sudo firewall-cmd --reload
```

9. Start and enable the `libvirtd` service.

```
[kni@provisioner ~]$ sudo systemctl start libvirtd
[kni@provisioner ~]$ sudo systemctl enable libvirtd --now
```

10. Create the `default` storage pool and start it.

```
[kni@provisioner ~]$ sudo virsh pool-define-as --name default --type dir --target /var/lib/libvirt/images
[kni@provisioner ~]$ sudo virsh pool-start default
[kni@provisioner ~]$ sudo virsh pool-autostart default
```

11. Configure networking.



This step can also be run from the web console.

Provisioning Network (IPv4 address)

```
[kni@provisioner ~]$ sudo nohup bash -c ""
nmcli con down "$PROV_CONN"
```



```
nmcli con delete "$PROV_CONN"
# RHEL 8.1 appends the word "System" in front of the connection, delete in case
it exists
nmcli con down "System $PROV_CONN"
nmcli con delete "System $PROV_CONN"
nmcli connection add ifname provisioning type bridge con-name provisioning
nmcli con add type bridge-slave ifname "$PROV_CONN" master provisioning
nmcli connection modify provisioning ipv4.addresses 172.22.0.1/24 ipv4.method
manual
nmcli con down provisioning
nmcli con up provisioning"""
```



The `ssh` connection might disconnect after executing this step.

The IPv4 address may be any address as long as it is not routable via the `baremetal` network.

Provisioning Network (IPv6 address)

```
[kni@provisioner ~]$ sudo nohup bash -c ""
nmcli con down "$PROV_CONN"
nmcli con delete "$PROV_CONN"
# RHEL 8.1 appends the word "System" in front of the connection, delete in case
it exists
nmcli con down "System $PROV_CONN"
nmcli con delete "System $PROV_CONN"
nmcli connection add ifname provisioning type bridge con-name provisioning
nmcli con add type bridge-slave ifname "$PROV_CONN" master provisioning
nmcli connection modify provisioning ipv6.addresses fd00:1101::1/64 ipv6.method
manual
nmcli con down provisioning
nmcli con up provisioning"""
```



The `ssh` connection might disconnect after executing this step.

The IPv6 address may be any address as long as it is not routable via the `baremetal` network.



Ensure that UEFI is enabled and UEFI PXE settings are set to the IPv6 protocol when using IPv6 addressing.

12. `ssh` back into the `provisioner` node (if required).

```
# ssh kni@provisioner.<cluster-name>.<domain>
```

13. Verify the connection bridges have been properly created.

```
[kni@provisioner ~]$ nmcli con show
```

NAME	UUID	TYPE	DEVICE
baremetal	4d5133a5-8351-4bb9-bfd4-3af264801530	bridge	baremetal
provisioning	43942805-017f-4d7d-a2c2-7cb3324482ed	bridge	provisioning
virbr0	d9bca40f-eee1-410b-8879-a2d4bb0465e7	bridge	virbr0
bridge-slave-eno1	76a8ed50-c7e5-4999-b4f6-6d9014dd0812	ethernet	eno1
bridge-slave-eno2	f31c3353-54b7-48de-893a-02d2b34c4736	ethernet	eno2

14. Create a `pull-secret.txt` file.

```
[kni@provisioner ~]$ vim pull-secret.txt
```

In a web browser, navigate to [Install on Bare Metal with user-provisioned infrastructure](#), and scroll down to the **Downloads** section. Click **Copy pull secret**. Paste the contents into the `pull-secret.txt` file and save the contents in the `kni` user's home directory.

3.3. Retrieving the OpenShift Container Platform installer (GA Release)

Use the `latest-4.x` version of the installer to deploy the latest generally available version of OpenShift Container Platform:

```
[kni@provisioner ~]$ export VERSION=latest-4.6
export RELEASE_IMAGE=$(curl -s https://mirror.openshift.com/pub/openshift-
v4/clients/ocp/$VERSION/release.txt | grep 'Pull From: quay.io' | awk -F ' ' '{print
$3}')
```

3.4. Extracting the OpenShift Container Platform installer (GA Release)

After retrieving the installer, the next step is to extract it.

Procedure

1. Set the environment variables:

```
[kni@provisioner ~]$ export cmd=openshift-baremetal-install
[kni@provisioner ~]$ export pullsecret_file=~/.pull-secret.txt
[kni@provisioner ~]$ export extract_dir=$(pwd)
```

2. Get the `oc` binary:

```
[kni@provisioner ~]$ curl -s https://mirror.openshift.com/pub/openshift-  
v4/clients/ocp/$VERSION/openshift-client-linux.tar.gz | tar zxvf - oc
```

3. Extract the installer:

```
[kni@provisioner ~]$ sudo cp oc /usr/local/bin  
[kni@provisioner ~]$ oc adm release extract --registry-config "${pullsecret_file}"  
--command=$cmd --to "${extract_dir}" ${RELEASE_IMAGE}  
[kni@provisioner ~]$ sudo cp openshift-baremetal-install /usr/local/bin
```

3.5. Creating an RHCOS images cache (optional)

To employ image caching, you must download two images: the Red Hat Enterprise Linux CoreOS (RHCOS) image used by the bootstrap VM and the RHCOS image used by the installer to provision the different nodes. Image caching is optional, but especially useful when running the installer on a network with limited bandwidth.

If you are running the installer on a network with limited bandwidth and the RHCOS images download takes more than 15 to 20 minutes, the installer will timeout. Caching images on a web server will help in such scenarios.

Use the following steps to install a container that contains the images.

1. Install **podman**.

```
$ sudo dnf install -y podman
```

2. Open firewall port **8080** to be used for RHCOS image caching.

```
$ sudo firewall-cmd --add-port=8080/tcp --zone=public --permanent  
$ sudo firewall-cmd --reload
```

3. Create a directory to store the **bootstrapimage** and **clusterosimage**.

```
$ mkdir /home/kni/rhcos_image_cache
```

4. Set the appropriate SELinux context for the newly created directory.

```
$ sudo semanage fcontext -a -t httpd_sys_content_t  
"/home/kni/rhcos_image_cache(/.*)?"  
$ sudo restorecon -Rv rhcos_image_cache/
```

5. Get the commit ID from the installer. The ID determines which images the installer needs to

download.

```
$ export COMMIT_ID=$(/usr/local/bin/openshift-baremetal-install version | grep  
'^built from commit' | awk '{print $4}')
```

6. Get the URI for the RHCOS image that the installer will deploy on the nodes.

```
$ export RHCOS_OPENSTACK_URI=$(curl -s -S  
https://raw.githubusercontent.com/openshift/installer/$COMMIT_ID/data/data/rhcos.js  
on | jq .images.openstack.path | sed 's/"//g')
```

7. Get the URI for the RHCOS image that the installer will deploy on the bootstrap VM.

```
$ export RHCOS_QEMU_URI=$(curl -s -S  
https://raw.githubusercontent.com/openshift/installer/$COMMIT_ID/data/data/rhcos.js  
on | jq .images.qemu.path | sed 's/"//g')
```

8. Get the path where the images are published.

```
$ export RHCOS_PATH=$(curl -s -S  
https://raw.githubusercontent.com/openshift/installer/$COMMIT_ID/data/data/rhcos.js  
on | jq .baseURI | sed 's/"//g')
```

9. Get the SHA hash for the RHCOS image that will be deployed on the bootstrap VM.

```
$ export RHCOS_QEMU_SHA_UNCOMPRESSED=$(curl -s -S  
https://raw.githubusercontent.com/openshift/installer/$COMMIT_ID/data/data/rhcos.js  
on | jq -r '.images.qemu["uncompressed-sha256"]')
```

10. Get the SHA hash for the RHCOS image that will be deployed on the nodes.

```
$ export RHCOS_OPENSTACK_SHA_COMPRESSED=$(curl -s -S  
https://raw.githubusercontent.com/openshift/installer/$COMMIT_ID/data/data/rhcos.js  
on | jq -r '.images.openstack.sha256')
```

11. Download the images and place them in the `/home/kni/rhcos_image_cache` directory.

```
$ curl -L ${RHCOS_PATH}${RHCOS_QEMU_URI} -o /home/kni/rhcos_image_cache/  
${RHCOS_QEMU_URI}  
$ curl -L ${RHCOS_PATH}${RHCOS_OPENSTACK_URI} -o /home/kni/rhcos_image_cache/  
${RHCOS_OPENSTACK_URI}
```

12. Confirm SELinux type is of `httpd_sys_content_t` for the newly created files.

```
$ ls -Z /home/kni/rhcos_image_cache
```

13. Create the pod.

```
$ podman run -d --name rhcos_image_cache \
-v /home/kni/rhcos_image_cache:/var/www/html \
-p 8080:8080/tcp \
registry.centos.org/centos/httpd-24-centos7:latest
```

14. Generate the `bootstrapOSImage` and `clusterOSImage` configuration.

```
$ export BAREMETAL_IP=$(ip addr show dev baremetal | awk '/inet /{print $2}' | cut
-d"/" -f1)
$ export RHCOS_OPENSTACK_SHA256=$(zcat /home/kni/rhcos_image_cache/
${RHCOS_OPENSTACK_URI} | sha256sum | awk '{print $1}')
$ export RHCOS_QEMU_SHA256=$(zcat /home/kni/rhcos_image_cache/${RHCOS_QEMU_URI} |
sha256sum | awk '{print $1}')
$ export CLUSTER_OS_IMAGE="http://${BAREMETAL_IP}:8080/${RHCOS_OPENSTACK_URI}
?sha256=${RHCOS_OPENSTACK_SHA256}"
$ export BOOTSTRAP_OS_IMAGE="http://${BAREMETAL_IP}:8080/${RHCOS_QEMU_URI}
?sha256=${RHCOS_QEMU_SHA256}"
$ echo "${RHCOS_OPENSTACK_SHA256} ${RHCOS_OPENSTACK_URI}" >
/home/kni/rhcos_image_cache/rhcos-ootpa-latest.qcow2.md5sum
$ echo "    bootstrapOSImage=${BOOTSTRAP_OS_IMAGE}"
$ echo "    clusterOSImage=${CLUSTER_OS_IMAGE}"
```

15. Add the required configuration to the `install-config.yaml` file under `platform.baremetal`.

```
platform:
  baremetal:
    bootstrapOSImage: http://<BAREMETAL_IP>:8080/<RHCOS_QEMU_URI>?sha256
=<RHCOS_QEMU_SHA256>
    clusterOSImage: http://<BAREMETAL_IP>:8080/<RHCOS_OPENSTACK_URI>?sha256
=<RHCOS_OPENSTACK_SHA256>
```

See the **Configuring the `install-config.yaml` file** section for additional details.

3.6. Configuration files

3.6.1. Configuring the `install-config.yaml` file

The `install-config.yaml` file requires some additional details. Most of the information is teaching the installer and the resulting cluster enough about the available hardware so that it is able to fully manage it.

1. Configure `install-config.yaml`. Change the appropriate variables to match the environment, including `pullSecret` and `sshKey`.

```
apiVersion: v1
basedomain: <domain>
metadata:
  name: <cluster-name>
networking:
  machineCIDR: <public-cidr>
  networkType: OVNKubernetes
compute:
- name: worker
  replicas: 2 ①
controlPlane:
  name: master
  replicas: 3
  platform:
    baremetal: {}
platform:
  baremetal:
    apiVIP: <api-ip>
    ingressVIP: <wildcard-ip>
    provisioningNetworkInterface: <NIC1>
    provisioningNetworkCIDR: <CIDR>
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: ipmi://<out-of-band-ip> ②
          username: <user>
          password: <password>
          bootMACAddress: <NIC1-mac-address>
          hardwareProfile: default
      - name: openshift-master-1
        role: master
        bmc:
          address: ipmi://<out-of-band-ip>
          username: <user>
          password: <password>
          bootMACAddress: <NIC1-mac-address>
          hardwareProfile: default
      - name: openshift-master-2
        role: master
        bmc:
          address: ipmi://<out-of-band-ip>
          username: <user>
          password: <password>
          bootMACAddress: <NIC1-mac-address>
          hardwareProfile: default
      - name: openshift-worker-0
```

```

    role: worker
    bmc:
      address: ipmi://<out-of-band-ip>
      username: <user>
      password: <password>
    bootMACAddress: <NIC1-mac-address>
    hardwareProfile: unknown
- name: openshift-worker-1
  role: worker
  bmc:
    address: ipmi://<out-of-band-ip>
    username: <user>
    password: <password>
  bootMACAddress: <NIC1-mac-address>
  hardwareProfile: unknown
pullSecret: '<pull_secret>'
sshKey: '<ssh_pub_key>'

```

- ① Scale the worker machines based on the number of worker nodes that are part of the OpenShift Container Platform cluster.
- ② Refer to the [BMC addressing](#) for more options

2. Create a directory to store cluster configs.

```

[kni@provisioner ~]$ mkdir ~/clusterconfigs
[kni@provisioner ~]$ cp install-config.yaml ~/clusterconfigs

```

3. Ensure all bare metal nodes are powered off prior to installing the OpenShift Container Platform cluster.

```

[kni@provisioner ~]$ ipmitool -I lanplus -U <user> -P <password> -H <management-server-ip> power off

```

4. Remove old bootstrap resources if any are left over from a previous deployment attempt.

```

for i in $(sudo virsh list | tail -n +3 | grep bootstrap | awk {'print $2'});
do
  sudo virsh destroy $i;
  sudo virsh undefine $i;
  sudo virsh vol-delete $i --pool $i;
  sudo virsh vol-delete $i.ign --pool $i;
  sudo virsh pool-destroy $i;
  sudo virsh pool-undefine $i;
done

```

3.6.2. Setting proxy settings within the `install-config.yaml` file (optional)

To deploy an OpenShift Container Platform cluster using a proxy, make the following changes to the `install-config.yaml` file.

```
apiVersion: v1
baseDomain: <domain>
proxy:
  httpProxy: http://USERNAME:PASSWORD@proxy.example.com:PORT
  httpsProxy: https://USERNAME:PASSWORD@proxy.example.com:PORT
  noProxy: <WILDCARD_OF_DOMAIN>,<PROVISIONING_NETWORK/CIDR>,<BMC_ADDRESS_RANGE/CIDR>
```

See below for an example of `noProxy` with values.

```
noProxy: .example.com,172.22.0.0/24,10.10.0.0/24
```

With a proxy enabled, set the appropriate values of the proxy in the corresponding key/value pair.

Key considerations:

- If the proxy does not have an HTTPS proxy, change the value of `httpsProxy` from `https://` to `http://`.
- If using a provisioning network, include it in the `noProxy` setting, otherwise the installer will fail.
- Set all of the proxy settings as environment variables within the provisioner node. For example, `HTTP_PROXY`, `HTTPS_PROXY`, and `NO_PROXY`.

3.6.3. Modifying the `install-config.yaml` file for no provisioning network (optional)

To deploy an OpenShift Container Platform cluster without a `provisioning` network, make the following changes to the `install-config.yaml` file.

```
platform:
  baremetal:
    apiVIP: <apiVIP>
    ingressVIP: <ingress/wildcard VIP>
    provisioningNetwork: "Disabled"
    provisioningHostIP: <baremetal_network_IP1>
    bootstrapProvisioningIP: <baremetal_network_IP2>
```



Requires providing two IP addresses from the `baremetal` network for the `provisioningHostIP` and `bootstrapProvisioningIP` configuration settings, and removing the `provisioningBridge` and `provisioningNetworkCIDR` configuration settings.

3.6.4. Additional `install-config` parameters

See the following tables for the required parameters, the `hosts` parameter, and the `bmc` parameter for the `install-config.yaml` file.

Table 2. Required parameters

Parameters	Default	Description
<code>baseDomain</code>		The domain name for the cluster. For example, <code>example.com</code> .
<code>bootMode</code>	<code>legacy</code>	The boot mode for a node. Options are <code>legacy</code> , <code>UEFI</code> and <code>UEFISecureBoot</code> .
<code>sshKey</code>		The <code>sshKey</code> configuration setting contains the key in the <code>~/.ssh/id_rsa.pub</code> file required to access the control plane nodes and worker nodes. Typically, this key is from the <code>provisioner</code> node.
<code>pullSecret</code>		The <code>pullSecret</code> configuration setting contains a copy of the pull secret downloaded from the Install OpenShift on Bare Metal page when preparing the provisioner node.
<pre>metadata: name:</pre>		The name to be given to the OpenShift Container Platform cluster. For example, <code>openshift</code> .
<pre>networking: machineCIDR:</pre>		The public CIDR (Classless Inter-Domain Routing) of the external network. For example, <code>10.0.0.0/24</code> or <code>2620:52:0:1302::/64</code> .
<pre>compute: - name: worker</pre>		The OpenShift Container Platform cluster requires a name be provided for worker (or compute) nodes even if there are zero nodes.
<pre>compute: replicas: 2</pre>		Replicas sets the number of worker (or compute) nodes in the OpenShift Container Platform cluster.

Parameters	Default	Description
<code>controlPlane: name: master</code>		The OpenShift Container Platform cluster requires a name for control plane (master) nodes.
<code>controlPlane: replicas: 3</code>		Replicas sets the number of control plane (master) nodes included as part of the OpenShift Container Platform cluster.
<code>provisioningNetworkInterface</code>		The name of the network interface on control plane nodes connected to the provisioning network.
<code>defaultMachinePlatform</code>		The default configuration used for machine pools without a platform configuration.
<code>apiVIP</code>	<code>api.<clustername>.clusterdomain></code>	The VIP to use for internal API communication. This setting must either be provided or pre-configured in the DNS so that the default name resolves correctly.
<code>disableCertificateVerification</code>	False	<code>redfish</code> and <code>redfish-virtualmedia</code> need this parameter to manage BMC addresses. The value should be <code>True</code> when using a self-signed certificate for BMC addresses.
<code>ingressVIP</code>	<code>test.apps.<clustername>.clusterdomain></code>	The VIP to use for ingress traffic.

Table 3. Optional Parameters

Parameters	Default	Description
<code>provisioningDHCPRange</code>	<code>172.22.0.10,172.22.0.100</code>	Defines the IP range for nodes on the <code>provisioning</code> network.
<code>provisioningNetworkCIDR</code>	<code>172.22.0.0/24</code>	The CIDR for the network to use for provisioning. This option is required when not using the default address range on the <code>provisioning</code> network.

Parameters	Default	Description
<code>clusterProvisioningIP</code>	The third IP address of the <code>provisioningNetworkCIDR</code> .	The IP address within the cluster where the provisioning services run. Defaults to the third IP address of the <code>provisioning</code> subnet. For example, <code>172.22.0.3</code> .
<code>bootstrapProvisioningIP</code>	The second IP address of the <code>provisioningNetworkCIDR</code> .	The IP address on the bootstrap VM where the provisioning services run while the installer is deploying the control plane (master) nodes. Defaults to the second IP address of the <code>provisioning</code> subnet. For example, <code>172.22.0.2</code> or <code>2620:52:0:1307::2</code>. Set this parameter to an available IP address on the <code>baremetal</code> network when the <code>provisioningNetwork</code> configuration setting is set to <code>Disabled</code>.
<code>externalBridge</code>	<code>baremetal</code>	The name of the <code>baremetal</code> bridge of the hypervisor attached to the <code>baremetal</code> network.
<code>provisioningBridge</code>	<code>provisioning</code>	The name of the <code>provisioning</code> bridge on the <code>provisioner</code> host attached to the <code>provisioning</code> network.
<code>defaultMachinePlatform</code>		The default configuration used for machine pools without a platform configuration.
<code>bootstrapOSImage</code>		A URL to override the default operating system image for the bootstrap node. The URL must contain a SHA-256 hash of the image. For example: <code>https://mirror.openshift.com/rhcos-&lt;version&gt;-qemu.qcow2.gz?sha256=&lt;uncompressed_sha256&gt;</code> or <code>http://[2620:52:0:1307::1]/rhcos-&lt;version&gt;-qemu.x86_64.qcow2.gz?sha256=&lt;uncompressed_sha256&gt;</code> .
<code>clusterOSImage</code>		A URL to override the default operating system for cluster nodes. The URL must include a SHA-256 hash of the image. For example, <code>https://mirror.openshift.com/images/rhcos-&lt;version&gt;-openstack.qcow2.gz?sha256=&lt;compressed_sha256&gt;</code> .

Parameters	Default	Description
<code>provisioningNetwork</code>		Set this parameter to <code>Disabled</code> to disable the requirement for a <code>provisioning</code> network. User may only do virtual media based provisioning, or bring up the cluster using assisted installation. If using power management, BMC's must be accessible from the machine networks. User must provide two IP addresses on the external network that are used for the provisioning services. Set this parameter to <code>Managed</code>, which is the default, to fully manage the provisioning network, including DHCP, TFTP, and so on. Set this parameter to <code>Unmanaged</code> to still enable the provisioning network but take care of manual configuration of DHCP. Virtual media provisioning is recommended but PXE is still available if required.
<code>provisioningHostIP</code>		Set this parameter to an available IP address on the <code>baremetal</code> network when the <code>provisioningNetwork</code> configuration setting is set to <code>Disabled</code> .
<code>httpProxy</code>		Set this parameter to the appropriate HTTP proxy used within your environment.
<code>httpsProxy</code>		Set this parameter to the appropriate HTTPS proxy used within your environment.
<code>noProxy</code>		Set this parameter to the appropriate list of exclusions for proxy usage within your environment.

Hosts

The `hosts` parameter is a list of separate bare metal assets used to build the cluster.

Name	Default	Description
<code>name</code>		The name of the <code>BareMetalHost</code> resource to associate with the details. For example, <code>openshift-master-0</code> .
<code>role</code>		The role of the bare metal node. Either <code>master</code> or <code>worker</code> .
<code>bmc</code>		Connection details for the baseboard management controller. See the BMC addressing section for additional details.
<code>bootMACAddress</code>		The MAC address of the NIC the host will use to boot on the <code>provisioning</code> network.

3.6.5. BMC addressing

Most vendors support BMC addressing with the Intelligent Platform Management Interface or IPMI. IPMI does not encrypt communications. It is suitable for use within a data center over a secured or dedicated management network. Check with your vendor to see if they support Redfish network boot. Redfish delivers simple and secure management for converged, hybrid IT and the Software Defined Data Center or SDDC. Redfish is human readable and machine capable, and leverages common Internet and web services standards to expose information directly to the modern tool chain. If your hardware does not support Redfish network boot, use IPMI.

IPMI

Hosts using IPMI use the `ipmi://<out-of-band-ip>:<port>` address format, which defaults to port `623` if not specified. The following example demonstrates an IPMI configuration within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: ipmi://<out-of-band-ip>
          username: <user>
          password: <password>
```

Redfish network boot

To enable Redfish, use `redfish://` or `redfish+http://` to disable TLS. The installer requires both the host name or the IP address and the path to the system ID. The following example demonstrates a Redfish configuration within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: redfish://<out-of-band-ip>/redfish/v1/Systems/1
          username: <user>
          password: <password>
```

While it is recommended to have a certificate of authority for the out-of-band management addresses, you must include `disableCertificateVerification: True` in the `bmc` configuration if using self-signed certificates. The following example demonstrates a Redfish configuration using the `disableCertificateVerification: True` configuration parameter within the `install-config.yaml` file.

```
platform:
  baremetal:
```

```

hosts:
  - name: openshift-master-0
    role: master
    bmc:
      address: redfish://<out-of-band-ip>/redfish/v1/Systems/1
      username: <user>
      password: <password>
      disableCertificateVerification: True

```

BMC addressing for Dell iDRAC

The `address` field for each `bmc` entry is a URL for connecting to the OpenShift Container Platform cluster nodes, including the type of controller in the URL scheme and its location on the network.

```

platform:
  baremetal:
    hosts:
      - name: <host name>
        role: <master | worker>
        bmc:
          address: <address> ①
          username: <user>
          password: <password>

```

① The `address` configuration setting specifies the protocol.

For Dell hardware, Red Hat supports integrated Dell Remote Access Controller (iDRAC) virtual media, Redfish network boot, and IPMI.

Table 4. BMC address formats for Dell iDRAC

Protocol	Address Format
iDRAC virtual media	<code>idrac-virtualmedia://<out-of-band-ip>/redfish/v1/Systems/System.Embedded.1</code>
Redfish network boot	<code>redfish://<out-of-band-ip>/redfish/v1/Systems/System.Embedded.1</code>
IPMI	<code>ipmi://<out-of-band-ip></code>



Use `idrac-virtualmedia` as the protocol for Redfish virtual media. `redfish-virtualmedia` will not work on Dell hardware. Dell's `idrac-virtualmedia` uses the Redfish standard with Dell's OEM extensions.

See the following sections for additional details.

Redfish virtual media for Dell iDRAC

For Redfish virtual media on Dell servers, use `idrac-virtualmedia://` in the `address` setting. Using `redfish-virtualmedia://` will not work.



Redfish virtual media on Dell servers has a known issue in OpenShift Container Platform 4.6, which will be resolved in a future release.

The following example demonstrates using iDRAC virtual media within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: idrac-virtualmedia://<out-of-band-
ip>/redfish/v1/Systems/System.Embedded.1
          username: <user>
          password: <password>
```

While it is recommended to have a certificate of authority for the out-of-band management addresses, you must include `disableCertificateVerification: True` in the `bmc` configuration if using self-signed certificates. The following example demonstrates a Redfish configuration using the `disableCertificateVerification: True` configuration parameter within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: idrac-virtualmedia://<out-of-band-
ip>/redfish/v1/Systems/System.Embedded.1
          username: <user>
          password: <password>
          disableCertificateVerification: True
```



Currently, Redfish is only supported on Dell with iDRAC firmware versions 4.20.20.20 through 04.40.00.00 for installer-provisioned installations on bare metal deployments. There is a known issue with version 04.40.00.00. With iDRAC 9 firmware version 04.40.00.00, the Virtual Console plug-in defaults to `eHTML5`, which causes problems with the **InsertVirtualMedia** workflow. Set the plug-in to `HTML5` to avoid this issue. The menu path is: **Configuration** → **Virtual console** → **Plug-in Type** → **HTML5**.

Ensure the OpenShift Container Platform cluster nodes have AutoAttach Enabled through the iDRAC console. The menu path is: **Configuration** → **Virtual Media** → **Attach Mode** → **AutoAttach**.

Use `idrac-virtualmedia://` as the protocol for Redfish virtual media. Using `redfish-virtualmedia://` will not work on Dell hardware, because the `idrac-`

`virtualmedia://` protocol corresponds to the `idrac` hardware type and the Redfish protocol in Ironi. Dell's `idrac-virtualmedia://` protocol uses the Redfish standard with Dell's OEM extensions. Ironi also supports the `idrac` type with the WSMAN protocol. Therefore, you must specify `idrac-virtualmedia://` to avoid unexpected behavior when electing to use Redfish with virtual media on Dell hardware.

Redfish network boot for iDRAC

To enable Redfish, use `redfish://` or `redfish+http://` to disable transport layer security (TLS). The installer requires both the host name or the IP address and the path to the system ID. The following example demonstrates a Redfish configuration within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: redfish://<out-of-band-ip>/redfish/v1/Systems/System.Embedded.1
          username: <user>
          password: <password>
```

While it is recommended to have a certificate of authority for the out-of-band management addresses, you must include `disableCertificateVerification: True` in the `bmc` configuration if using self-signed certificates. The following example demonstrates a Redfish configuration using the `disableCertificateVerification: True` configuration parameter within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: redfish://<out-of-band-ip>/redfish/v1/Systems/System.Embedded.1
          username: <user>
          password: <password>
          disableCertificateVerification: True
```



Currently, Redfish is only supported on Dell hardware with iDRAC firmware versions `4.20.20.20` through `04.40.00.00` for installer-provisioned installations on bare metal deployments. There is a known issue with version `04.40.00.00`. With iDRAC 9 firmware version `04.40.00.00`, the Virtual Console plug-in defaults to `eHTML5`, which causes problems with the **InsertVirtualMedia** workflow. Set the plug-in to `HTML5` to avoid this issue. The menu path is: **Configuration** → **Virtual console** → **Plug-in Type** → **HTML5**.

Ensure the OpenShift Container Platform cluster nodes have AutoAttach Enabled through the iDRAC console. The menu path is: **Configuration** → **Virtual Media** →

Attach Mode → AutoAttach .

The `redfish://` URL protocol corresponds to the `redfish` hardware type in Ironic.

BMC addressing for HPE iLO

The `address` field for each `bmc` entry is a URL for connecting to the OpenShift Container Platform cluster nodes, including the type of controller in the URL scheme and its location on the network.

```
platform:
  baremetal:
    hosts:
      - name: <host name>
        role: <master | worker>
        bmc:
          address: <address> ①
          username: <user>
          password: <password>
```

① The `address` configuration setting specifies the protocol.

For HPE integrated Lights Out (iLO), Red Hat supports Redfish virtual media, Redfish network boot, and IPMI.

Table 5. BMC address formats for HPE iLO

Protocol	Address Format
Redfish virtual media	<code>redfish-virtualmedia://<out-of-band-ip>/redfish/v1/Systems/1</code>
Redfish network boot	<code>redfish://<out-of-band-ip>/redfish/v1/Systems/1</code>
IPMI	<code>ipmi://<out-of-band-ip></code>

See the following sections for additional details.

Redfish virtual media for HPE iLO

To enable Redfish virtual media for HPE servers, use `redfish-virtualmedia://` in the `address` setting. The following example demonstrates using Redfish virtual media within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: redfish-virtualmedia://<out-of-band-ip>/redfish/v1/Systems/1
          username: <user>
```

```
password: <password>
```

While it is recommended to have a certificate of authority for the out-of-band management addresses, you must include `disableCertificateVerification: True` in the `bmc` configuration if using self-signed certificates. The following example demonstrates a Redfish configuration using the `disableCertificateVerification: True` configuration parameter within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: redfish-virtualmedia://<out-of-band-ip>/redfish/v1/Systems/1
          username: <user>
          password: <password>
          disableCertificateVerification: True
```



Redfish virtual media is not supported on 9th generation systems running iLO4, because IroniC does not support iLO4 with virtual media.

Redfish network boot for HPE iLO

To enable Redfish, use `redfish://` or `redfish+http://` to disable TLS. The installer requires both the host name or the IP address and the path to the system ID. The following example demonstrates a Redfish configuration within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
        bmc:
          address: redfish://<out-of-band-ip>/redfish/v1/Systems/1
          username: <user>
          password: <password>
```

While it is recommended to have a certificate of authority for the out-of-band management addresses, you must include `disableCertificateVerification: True` in the `bmc` configuration if using self-signed certificates. The following example demonstrates a Redfish configuration using the `disableCertificateVerification: True` configuration parameter within the `install-config.yaml` file.

```
platform:
  baremetal:
    hosts:
      - name: openshift-master-0
        role: master
```

```
bmc:
  address: redfish://<out-of-band-ip>/redfish/v1/Systems/1
  username: <user>
  password: <password>
  disableCertificateVerification: True
```

3.6.6. Root device hints

The `rootDeviceHints` parameter enables the installer to provision the Red Hat Enterprise Linux CoreOS (RHCOS) image to a particular device. The installer examines the devices in the order it discovers them, and compares the discovered values with the hint values. The installer uses the first discovered device that matches the hint value. The configuration can combine multiple hints, but a device must match all hints for the installer to select it.

Table 6. Subfields

Subfield	Description
<code>deviceName</code>	A string containing a Linux device name like <code>/dev/vda</code> . The hint must match the actual value exactly.
<code>hctl</code>	A string containing a SCSI bus address like <code>0:0:0:0</code> . The hint must match the actual value exactly.
<code>model</code>	A string containing a vendor-specific device identifier. The hint can be a substring of the actual value.
<code>vendor</code>	A string containing the name of the vendor or manufacturer of the device. The hint can be a sub-string of the actual value.
<code>serialNumber</code>	A string containing the device serial number. The hint must match the actual value exactly.
<code>minSizeGigabytes</code>	An integer representing the minimum size of the device in gigabytes.
<code>wwn</code>	A string containing the unique storage identifier. The hint must match the actual value exactly.
<code>wwnWithExtension</code>	A string containing the unique storage identifier with the vendor extension appended. The hint must match the actual value exactly.
<code>wwnVendorExtension</code>	A string containing the unique vendor storage identifier. The hint must match the actual value exactly.
<code>rotational</code>	A Boolean indicating whether the device should be a rotating disk (true) or not (false).

Example usage

```
- name: master-0
  role: master
  bmc:
    address: ipmi://10.10.0.3:6203
    username: admin
    password: redhat
  bootMACAddress: de:ad:be:ef:00:40
  rootDeviceHints:
    deviceName: "/dev/sda"
```

3.6.7. Creating the OpenShift Container Platform manifests

1. Create the OpenShift Container Platform manifests.

```
[kni@provisioner ~]$ ./openshift-baremetal-install --dir ~/clusterconfigs create
manifests
```

```
INFO Consuming Install Config from target directory
WARNING Making control-plane schedulable by setting MastersSchedulable to true for
Scheduler cluster settings
WARNING Discarding the Openshift Manifest that was provided in the target directory
because its dependencies are dirty and it needs to be regenerated
```

3.7. Creating a disconnected registry (optional)

In some cases, you might want to install an OpenShift Container Platform cluster using a local copy of the installation registry. This could be for enhancing network efficiency because the cluster nodes are on a network that does not have access to the internet.

A local, or mirrored, copy of the registry requires the following:

- A certificate for the registry node. This can be a self-signed certificate.
- A web server that a container on a system will serve.
- An updated pull secret that contains the certificate and local repository information.



Creating a disconnected registry on a registry node is optional. The subsequent sections indicate that they are optional since they are steps you need to execute only when creating a disconnected registry on a registry node. You should execute all of the subsequent sub-sections labeled "(optional)" when creating a disconnected registry on a registry node.

3.7.1. Preparing the registry node to host the mirrored registry (optional)

Make the following changes to the registry node.

Procedure

1. Open the firewall port on the registry node.

```
[user@registry ~]$ sudo firewall-cmd --add-port=5000/tcp --zone=libvirt
--permanent
[user@registry ~]$ sudo firewall-cmd --add-port=5000/tcp --zone=public
--permanent
[user@registry ~]$ sudo firewall-cmd --reload
```

2. Install the required packages for the registry node.

```
[user@registry ~]$ sudo yum -y install python3 podman httpd httpd-tools jq
```

3. Create the directory structure where the repository information will be held.

```
[user@registry ~]$ sudo mkdir -p /opt/registry/{auth,certs,data}
```

3.7.2. Generating the self-signed certificate (optional)

Generate a self-signed certificate for the registry node and put it in the `/opt/registry/certs` directory.

Procedure

1. Adjust the certificate information as appropriate.

```
[user@registry ~]$ host_fqdn=$( hostname --long )
[user@registry ~]$ cert_c="<Country Name>" # Country Name (C, 2 letter code)
[user@registry ~]$ cert_s="<State>" # Certificate State (S)
[user@registry ~]$ cert_l="<Locality>" # Certificate Locality (L)
[user@registry ~]$ cert_o="<Organization>" # Certificate Organization (O)
[user@registry ~]$ cert_ou="<Org Unit>" # Certificate Organizational Unit (OU)
[user@registry ~]$ cert_cn="${host_fqdn}" # Certificate Common Name (CN)

[user@registry ~]$ openssl req \
    -newkey rsa:4096 \
    -nodes \
    -sha256 \
    -keyout /opt/registry/certs/domain.key \
    -x509 \
    -days 365 \
    -out /opt/registry/certs/domain.crt \
    -addext "subjectAltName = DNS:${host_fqdn}" \
```

```
-subj "/C=${cert_c}/ST=${cert_s}/L=${cert_l}/O=${cert_o}/OU=${cert_ou}/CN=${cert_cn}"
```



When replacing **<Country Name>**, ensure that it only contains two letters. For example, **US**.

2. Update the registry node's **ca-trust** with the new certificate.

```
[user@registry ~]$ sudo cp /opt/registry/certs/domain.crt /etc/pki/ca-trust/source/anchors/
[user@registry ~]$ sudo update-ca-trust extract
```

3.7.3. Creating the registry podman container (optional)

The registry container uses the **/opt/registry** directory for certificates, authentication files, and to store its data files.

The registry container uses **httpd** and needs an **htpasswd** file for authentication.

Procedure

1. Create an **htpasswd** file in **/opt/registry/auth** for the container to use.

```
[user@registry ~]$ htpasswd -bBc /opt/registry/auth/htpasswd <user> <passwd>
```

Replace **<user>** with the user name and **<passwd>** with the password.

2. Create and start the registry container.

```
[user@registry ~]$ podman create \
  --name ocpdiscon-registry \
  -p 5000:5000 \
  -e "REGISTRY_AUTH=htpasswd" \
  -e "REGISTRY_AUTH_HTPASSWD_REALM=Registry" \
  -e "REGISTRY_HTTP_SECRET=ALongRandomSecretForRegistry" \
  -e "REGISTRY_AUTH_HTPASSWD_PATH=/auth/htpasswd" \
  -e "REGISTRY_HTTP_TLS_CERTIFICATE=/certs/domain.crt" \
  -e "REGISTRY_HTTP_TLS_KEY=/certs/domain.key" \
  -e "REGISTRY_COMPATIBILITY_SCHEMA1_ENABLED=true" \
  -v /opt/registry/data:/var/lib/registry:z \
  -v /opt/registry/auth:/auth:z \
  -v /opt/registry/certs:/certs:z \
  docker.io/library/registry:2
```

```
[user@registry ~]$ podman start ocpdiscon-registry
```

3.7.4. Copy and update the pull-secret (optional)

Copy the pull secret file from the provisioner node to the registry node and modify it to include the authentication information for the new registry node.

Procedure

1. Copy the `pull-secret.txt` file.

```
[user@registry ~]$ scp kni@provisioner:/home/kni/pull-secret.txt pull-secret.txt
```

2. Update the `host_fqdn` environment variable with the fully qualified domain name of the registry node.

```
[user@registry ~]$ host_fqdn=$( hostname --long )
```

3. Update the `b64auth` environment variable with the base64 encoding of the `http` credentials used to create the `htpasswd` file.

```
[user@registry ~]$ b64auth=$( echo -n '<username>:<passwd>' | openssl base64 )
```

Replace `<username>` with the user name and `<passwd>` with the password.

4. Set the `AUTHSTRING` environment variable to use the `base64` authorization string. The `$USER` variable is an environment variable containing the name of the current user.

```
[user@registry ~]$ AUTHSTRING="{\"$host_fqdn:5000\": {\"auth\": \"$b64auth\",  
\"email\": \"$USER@redhat.com\"}}\""
```

5. Update the `pull-secret.txt` file.

```
[user@registry ~]$ jq ".auths += $AUTHSTRING" < pull-secret.txt > pull-secret-update.txt
```

3.7.5. Mirroring the repository (optional)

Procedure

1. Copy the `oc` binary from the provisioner node to the registry node.

```
[user@registry ~]$ sudo scp kni@provisioner:/usr/local/bin/oc /usr/local/bin
```

2. Get the release image and mirror the remote install images to the local repository.

```
[user@registry ~]$ export VERSION=latest-4.6
[user@registry ~]$ UPSTREAM_REPO=$(curl -s
https://mirror.openshift.com/pub/openshift-
v4/x86_64/clients/ocp/$VERSION/release.txt | awk '/Pull From/ {print $3}')
[user@registry ~]$ /usr/local/bin/oc adm release mirror \
  -a pull-secret-update.txt
  --from=$UPSTREAM_REPO \
  --to-release-image=$LOCAL_REG/$LOCAL_REPO:${VERSION} \
  --to=$LOCAL_REG/$LOCAL_REPO
```

3.7.6. Modify the `install-config.yaml` file to use the disconnected registry (optional)

On the provisioner node, the `install-config.yaml` file should use the newly created pull-secret from the `pull-secret-update.txt` file. The `install-config.yaml` file must also contain the disconnected registry node's certificate and registry information.

Procedure

1. Add the disconnected registry node's certificate to the `install-config.yaml` file. The certificate should follow the `"additionalTrustBundle: |"` line and be properly indented, usually by two spaces.

```
$ echo "additionalTrustBundle: |" >> install-config.yaml
$ sed -e 's/^/  /' /opt/registry/certs/domain.crt >> install-config.yaml
```

2. Add the mirror information for the registry to the `install-config.yaml` file.

```
$ cat <<EOF >> install-config.yaml
imageContentSources:
- mirrors:
  - registry.example.com:5000/ocp4/openshift4
  source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
- mirrors:
  - registry.example.com:5000/ocp4/openshift4
  source: registry.svc.ci.openshift.org/ocp/release
- mirrors:
  - registry.example.com:5000/ocp4/openshift4
  source: quay.io/openshift-release-dev/ocp-release
EOF
```



Replace `registry.example.com` with the registry's fully qualified domain name.

3.8. Deploying routers on worker nodes

During installation, the installer deploys router pods on worker nodes. By default, the installer

installs two router pods. If the initial cluster has only one worker node, or if a deployed cluster requires additional routers to handle external traffic loads destined for services within the OpenShift Container Platform cluster, you can create a `yaml` file to set an appropriate number of router replicas.



By default, the installer deploys two routers. If the cluster has at least two worker nodes, you can skip this section. For more information on the Ingress Operator see: [Ingress Operator in OpenShift Container Platform](#).



If the cluster has no worker nodes, the installer deploys the two routers on the control plane nodes by default. If the cluster has no worker nodes, you can skip this section.

Procedure

1. Create a `router-replicas.yaml` file.

```
apiVersion: operator.openshift.io/v1
kind: IngressController
metadata:
  name: default
  namespace: openshift-ingress-operator
spec:
  replicas: <num-of-router-pods>
  endpointPublishingStrategy:
    type: HostNetwork
  nodePlacement:
    nodeSelector:
      matchLabels:
        node-role.kubernetes.io/worker: ""
```



Replace `<num-of-router-pods>` with an appropriate value. If working with just one worker node, set `replicas` to `1`. If working with more than 3 worker nodes, you can increase `replicas` from the default value `2` as appropriate.

2. Save and copy the `router-replicas.yaml` file to the `clusterconfigs/openshift` directory.

```
cp ~/router-replicas.yaml clusterconfigs/openshift/99_router-replicas.yaml
```

3.9. Validation checklist for installation

- ☐ OpenShift Container Platform installer has been retrieved.
- ☐ OpenShift Container Platform installer has been extracted.
- ☐ Required parameters for the `install-config.yaml` have been configured.
- ☐ The `hosts` parameter for the `install-config.yaml` has been configured.

- ☐ The `bmc` parameter for the `install-config.yaml` has been configured.
- ☐ Conventions for the values configured in the `bmc address` field have been applied.
- ☐ Created a disconnected registry (optional).
- ☐ Validate disconnected registry settings if in use. (optional)
- ☐ Deployed routers on worker nodes. (optional)

3.10. Deploying the cluster via the OpenShift Container Platform installer

Run the OpenShift Container Platform installer:

```
[kni@provisioner ~]$ ./openshift-baremetal-install --dir ~/clusterconfigs --log-level debug create cluster
```

3.11. Following the installation

During the deployment process, you can check the installation's overall status by issuing the `tail` command to the `.openshift_install.log` log file in the `install` directory folder.

```
[kni@provisioner ~]$ tail -f /path/to/install-dir/.openshift_install.log
```

Chapter 4. Day 2 operations

The following sections are optional, but may be of interest after the initial deployment has been completed.

4.1. Accessing the web console

The web console runs as a pod on the master. The static assets required to run the web console are served by the pod. Once OpenShift Container Platform is successfully installed, find the URL for the web console and login credentials for your installed cluster in the CLI output of the installation program. For example:

Example output

```
INFO Install complete!
INFO Run 'export KUBECONFIG=<your working directory>/auth/kubeconfig' to manage the
cluster with 'oc', the OpenShift CLI.
INFO The cluster is ready when 'oc login -u kubeadmin -p <provided>' succeeds (wait a
few minutes).
INFO Access the OpenShift web-console here: https://console-openshift-
console.apps.demo1.openshift4-beta-abcorp.com
INFO Login to the console with user: kubeadmin, password: <provided>
```

Use those details to log in and access the web console.

Additionally, you can execute:

```
oc whoami --show-console
```

To obtain the url for the console.

4.2. Backing up the cluster configuration

At this point you have a working OpenShift 4 cluster on baremetal. In order to take advantage of the baremetal hardware that was the provision node, you can repurpose the provisioning node as a worker. Prior to reprovisioning the node, it is recommended to backup some existing files.

Procedure

1. Tar the `clusterconfig` folder and download it to your local machine.

```
tar cvfz clusterconfig.tar.gz ~/clusterconfig
```

2. Copy the Private part for the SSH Key configured on the `install-config.yaml` file to your local machine.

```
tar cvfz clusterconfigsh.tar.gz ~/.ssh/id_rsa*
```

3. Copy the `install-config.yaml` and `metal3-config.yaml` files.

```
tar cvfz yamlconfigs.tar.gz install-config.yaml metal3-config.yaml
```

4.3. Expanding the cluster

After deploying an installer-provisioned OpenShift Container Platform cluster, you can use the following procedures to expand the number of worker nodes. Ensure that each prospective worker node meets the prerequisites.



Expanding the cluster using RedFish Virtual Media involves meeting minimum firmware requirements. See **Firmware requirements for installing with virtual media** in the **Prerequisites** section for additional details when expanding the cluster using RedFish Virtual Media.

4.3.1. Preparing the bare metal node

Expanding the cluster requires a DHCP server. Each node must have a DHCP reservation.

Preparing the bare metal node requires executing the following procedure from the provisioner node.

Procedure

1. Get the `oc` binary, if needed. It should already exist on the provisioner node.

```
[kni@provisioner ~]$ export VERSION=latest-4.6  
[kni@provisioner ~]$ curl -s https://mirror.openshift.com/pub/openshift-  
v4/clients/ocp/$VERSION/openshift-client-linux-$VERSION.tar.gz | tar zxvf - oc
```

```
[kni@provisioner ~]$ sudo cp oc /usr/local/bin
```

2. Power off the bare metal node via the baseboard management controller and ensure it is off.
3. Retrieve the user name and password of the bare metal node's baseboard management controller. Then, create `base64` strings from the user name and password. In the following example, the user name is `root` and the password is `calvin`.

```
[kni@provisioner ~]$ echo -ne "root" | base64
```

```
[kni@provisioner ~]$ echo -ne "calvin" | base64
```

4. Create a configuration file for the bare metal node.

```
[kni@provisioner ~]$ vim bmh.yaml
```

```
---
apiVersion: v1
kind: Secret
metadata:
  name: openshift-worker-<num>-bmc-secret
type: Opaque
data:
  username: <base64-of-uid>
  password: <base64-of-pwd>
---
apiVersion: metal3.io/v1alpha1
kind: BareMetalHost
metadata:
  name: openshift-worker-<num>
spec:
  online: true
  bootMACAddress: <NIC1-mac-address>
  bmc:
    address: <protocol>://<bmc-ip>
    credentialsName: openshift-worker-<num>-bmc-secret
```

Replace `<num>` for the worker number of the bare metal node in the two `name` fields and the `credentialsName` field. Replace `<base64-of-uid>` with the `base64` string of the user name. Replace `<base64-of-pwd>` with the `base64` string of the password. Replace `<NIC1-mac-address>` with the MAC address of the bare metal node's first NIC.

Refer to the BMC addressing section for additional BMC configuration options. Replace `<protocol>` with the BMC protocol, such as IPMI, RedFish, or others. Replace `<bmc-ip>` with the IP address of the bare metal node's baseboard management controller.



If the MAC address of an existing bare metal node matches the MAC address of a bare metal host that you are attempting to provision, then the Ironi installation will fail. If the host enrollment, inspection, cleaning, or other Ironi steps fail, the `metal3-baremetal-operator` will continuously retry. See [Diagnosing a host duplicate MAC address](#) for more information.

5. Create the bare metal node.

```
[kni@provisioner ~]$ oc -n openshift-machine-api create -f bmh.yaml
```

```
secret/openshift-worker-<num>-bmc-secret created
```

```
baremetalhost.metal3.io/openshift-worker-<num> created
```

Where `<num>` will be the worker number.

6. Power up and inspect the bare metal node.

```
[kni@provisioner ~]$ oc -n openshift-machine-api get bmh openshift-worker-<num>
```

Where `<num>` is the worker node number.

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
HARDWARE PROFILE	ONLINE	ERROR		
openshift-worker-<num>	OK	ready		ipmi://<out-of-
band-ip>	unknown	true		

4.3.2. Preparing to deploy with Virtual Media on the baremetal network

If the `provisioning` network is enabled, and you want to expand the cluster using Virtual Media on the `baremetal` network, execute the following procedure.

Procedure

1. Edit the provisioning configuration resource (CR) to enable deploying with Virtual Media on the `baremetal` network.

```
oc edit provisioning
```

```
apiVersion: metal3.io/v1alpha1
kind: Provisioning
metadata:
  creationTimestamp: "2021-08-05T18:51:50Z"
  finalizers:
  - provisioning.metal3.io
  generation: 8
  name: provisioning-configuration
  resourceVersion: "551591"
  uid: f76e956f-24c6-4361-aa5b-feaf72c5b526
spec:
  preProvisioningOSDownloadURLs: {}
  provisioningDHCPRange: 172.22.0.10,172.22.0.254
  provisioningIP: 172.22.0.3
  provisioningInterface: enp1s0
  provisioningNetwork: Managed
  provisioningNetworkCIDR: 172.22.0.0/24
  provisioningOSDownloadURL: http://192.168.111.1/images/rhcos-
49.84.202107010027-0-
openstack.x86_64.qcow2.gz?sha256=c7dde5f96826c33c97b5a4ad34110212281916128ae1110095
```

```
6f400db3d5299e
  virtualMediaViaExternalNetwork: true ①
status:
  generations:
  - group: apps
    hash: ""
    lastGeneration: 7
    name: metal3
    namespace: openshift-machine-api
    resource: deployments
  - group: apps
    hash: ""
    lastGeneration: 1
    name: metal3-image-cache
    namespace: openshift-machine-api
    resource: daemonsets
  observedGeneration: 8
  readyReplicas: 0
```

① Add `virtualMediaViaExternalNetwork: true` to the provisioning CR.

2. Edit the machine set to use the API VIP address.

```
oc edit machineset
```

```
apiVersion: machine.openshift.io/v1beta1
kind: MachineSet
metadata:
  creationTimestamp: "2021-08-05T18:51:52Z"
  generation: 11
  labels:
    machine.openshift.io/cluster-api-cluster: otest-hwmdt
    machine.openshift.io/cluster-api-machine-role: worker
    machine.openshift.io/cluster-api-machine-type: worker
  name: otest-hwmdt-worker-0
  namespace: openshift-machine-api
  resourceVersion: "551513"
  uid: fad1c6e0-b9da-4d4a-8d73-286f78788931
spec:
  replicas: 2
  selector:
    matchLabels:
      machine.openshift.io/cluster-api-cluster: otest-hwmdt
      machine.openshift.io/cluster-api-machineset: otest-hwmdt-worker-0
  template:
    metadata:
      labels:
        machine.openshift.io/cluster-api-cluster: otest-hwmdt
        machine.openshift.io/cluster-api-machine-role: worker
```

```

machine.openshift.io/cluster-api-machine-type: worker
machine.openshift.io/cluster-api-machineset: otest-hwmdt-worker-0
spec:
  metadata: {}
  providerSpec:
    value:
      apiVersion: baremetal.cluster.k8s.io/v1alpha1
      hostSelector: {}
      image:
        checksum: http://172.22.0.3:6181/images/rhcos-49.84.202107010027-0-
openstack.x86_64.qcow2/cached-rhcos-49.84.202107010027-0-
openstack.x86_64.qcow2.md5sum ①
        url: http://172.22.0.3:6181/images/rhcos-49.84.202107010027-0-
openstack.x86_64.qcow2/cached-rhcos-49.84.202107010027-0-openstack.x86_64.qcow2 ②
      kind: BareMetalMachineProviderSpec
      metadata:
        creationTimestamp: null
      userData:
        name: worker-user-data
  status:
    availableReplicas: 2
    fullyLabeledReplicas: 2
    observedGeneration: 11
    readyReplicas: 2
    replicas: 2

```

① Edit the `checksum` URL to use the API VIP address.

② Edit the `url` URL to use the API VIP address.

Diagnosing a duplicate MAC address when provisioning a new host in the cluster

If the MAC address of an existing bare-metal node in the cluster matches the MAC address of a bare-metal host you are attempting to add to the cluster, the Bare Metal Operator associates the host with the existing node. If the host enrollment, inspection, cleaning, or other IroniC steps fail, the Bare Metal Operator retries the installation continuously. A registration error is displayed for the failed bare-metal host.

You can diagnose a duplicate MAC address by examining the bare-metal hosts that are running in the `openshift-machine-api` namespace.

Prerequisites

- Install an OpenShift Container Platform cluster on bare metal.
- Install the OpenShift Container Platform CLI `oc`.
- Log in as a user with `cluster-admin` privileges.

Procedure

To determine whether a bare-metal host that fails provisioning has the same MAC address as an existing node, do the following:

1. Get the bare-metal hosts running in the `openshift-machine-api` namespace:

```
$ oc get bmh -n openshift-machine-api
```

Example output

NAME	STATUS	PROVISIONING STATUS	CONSUMER
openshift-master-0	OK	externally provisioned	openshift-zpwpq-master-0
openshift-master-1	OK	externally provisioned	openshift-zpwpq-master-1
openshift-master-2	OK	externally provisioned	openshift-zpwpq-master-2
openshift-worker-0-lv84n	OK	provisioned	openshift-zpwpq-worker-0-
openshift-worker-1-zd8lm	OK	provisioned	openshift-zpwpq-worker-0-
openshift-worker-2	error	registering	

2. To see more detailed information about the status of the failing host, run the following command replacing `<bare_metal_host_name>` with the name of the host:

```
$ oc get -n openshift-machine-api bmh <bare_metal_host_name> -o yaml
```

Example output

```
...
status:
  errorCount: 12
  errorMessage: MAC address b4:96:91:1d:7c:20 conflicts with existing node
openshift-worker-1
  errorType: registration error
...
```

4.3.3. Provisioning the bare metal node

Provisioning the bare metal node requires executing the following procedure from the provisioner node.

Procedure

1. Ensure the `PROVISIONING STATUS` is `ready` before provisioning the bare metal node.

```
[kni@provisioner ~]$ oc -n openshift-machine-api get bmh openshift-worker-<num>
```

Where `<num>` is the worker node number.

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
HARDWARE PROFILE	ONLINE	ERROR		

```
openshift-worker-<num>    OK      ready      ipmi://<out-of-
band-ip>    unknown    true
```

2. Get a count of the number of worker nodes.

```
[kni@provisioner ~]$ oc get nodes
```

NAME VERSION	STATUS	ROLES	AGE
provisioner.openshift.example.com v1.16.2	Ready	master	30h
openshift-master-1.openshift.example.com v1.16.2	Ready	master	30h
openshift-master-2.openshift.example.com v1.16.2	Ready	master	30h
openshift-master-3.openshift.example.com v1.16.2	Ready	master	30h
openshift-worker-0.openshift.example.com v1.16.2	Ready	master	30h
openshift-worker-1.openshift.example.com v1.16.2	Ready	master	30h

3. Get the machine set.

```
[kni@provisioner ~]$ oc get machinesets -n openshift-machine-api
```

NAME	DESIRED	CURRENT	READY	AVAILABLE	AGE
...					
openshift-worker-0.example.com	1	1	1	1	55m
openshift-worker-1.example.com	1	1	1	1	55m

4. Increase the number of worker nodes by one.

```
[kni@provisioner ~]$ oc scale --replicas=<num> machineset <machineset> -n
openshift-machine-api
```

Replace **<num>** with the new number of worker nodes. Replace **<machineset>** with the name of the machine set from the previous step.

5. Check the status of the bare metal node.

```
[kni@provisioner ~]$ oc -n openshift-machine-api get bmh openshift-worker-<num>
```

Where `<num>` is the worker node number. The status changes from `ready` to `provisioning`.

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
hardware profile	online	error		
openshift-worker- <code><num></code>	OK	provisioning	openshift-worker- <code><num></code>	
65tjz	ipmi:// <code><out-of-band-ip></code>	unknown	true	

The `provisioning` status remains until the OpenShift Container Platform cluster provisions the node. This can take 30 minutes or more. Once complete, the status will change to `provisioned`.

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
hardware profile	online	error		
openshift-worker- <code><num></code>	OK	provisioned	openshift-worker- <code><num></code>	
65tjz	ipmi:// <code><out-of-band-ip></code>	unknown	true	

6. Once provisioned, ensure the bare metal node is ready.

```
[kni@provisioner ~]$ oc get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
provisioner.openshift.example.com	Ready	master	30h	v1.16.2
openshift-master-1.openshift.example.com	Ready	master	30h	v1.16.2
openshift-master-2.openshift.example.com	Ready	master	30h	v1.16.2
openshift-master-3.openshift.example.com	Ready	master	30h	v1.16.2
openshift-worker-0.openshift.example.com	Ready	master	30h	v1.16.2
openshift-worker-1.openshift.example.com	Ready	master	30h	v1.16.2
openshift-worker- <code><num></code> .openshift.example.com	Ready	worker	3m27s	v1.16.2

You can also check the kubelet.

```
[kni@provisioner ~]$ ssh openshift-worker-<num>
```

```
[kni@openshift-worker-<num>]$ journalctl -fu kubelet
```

4.3.4. Preparing the provisioner node to be deployed as a worker node

Procedure

Perform the following steps prior to converting the provisioner node to a worker node.

1. `ssh` to a system (for example, a laptop) that can access the out of band management network of the current provisioner node.
2. Copy the backups `clusterconfig.tar.gz`, `clusterconfigsh.tar.gz`, and `amlconfigs.tar.gz` to the

new system.

3. Copy the `oc` binary from the existing provisioning node to the new system.
4. Make a note of the mac addresses, the baremetal network IP used for the provisioner node, and the IP address of the Out of band Management Network.
5. Reboot the system and ensure that PXE is enabled on the provisioning network and PXE is disabled for all other NICs.
6. If installation was performed using a Satellite server, remove the Host entry for the existing provisioning node.
7. Install the `ipmitool` on the new system in order to power off the provisioner node.

4.3.5. Adding a worker node to an existing cluster

Procedure

1. Retrieve the username and password of the bare metal node's baseboard management controller. Then, create `base64` strings from the username and password. In the following example, the username is `root` and the password is `calvin`.

```
[kni@provisioner ~]$ echo -ne "root" | base64
```

```
[kni@provisioner ~]$ echo -ne "calvin" | base64
```

2. Create a configuration file for the bare metal node.

```
[kni@provisioner ~]$ vim bmh.yaml
```

```
---
apiVersion: v1
kind: Secret
metadata:
  name: openshift-worker-<num>-bmc-secret
type: Opaque
data:
  username: <base64-of-uid>
  password: <base64-of-pwd>
---
apiVersion: metal3.io/v1alpha1
kind: BareMetalHost
metadata:
  name: openshift-worker-<num>
spec:
  online: true
  bootMACAddress: <NIC1-mac-address>
  bmc:
```

```
address: ipmi://<bmc-ip>
credentialsName: openshift-worker-<num>-bmc-secret
```

Replace `<num>` for the worker number of bare metal node in two `name` fields and `credentialsName` field. Replace `<base64-of-uid>` with the `base64` string of the username. Replace `<base64-of-pwd>` with the `base64` string of the password. Replace `<NIC1-mac-address>` with the MAC address of the bare metal node's first NIC. Replace `<bmc-ip>` with the IP address of the bare metal node's baseboard management controller.



When using `redfish` or `redfish-virtualmedia`, add the appropriate addressing as described in the BMC addressing section. See [BMC addressing](#) for details.

1. Create the bare metal node.

```
[kni@provisioner ~]$ oc -n openshift-machine-api create -f bmh.yaml
```

```
secret/openshift-worker-<num>-bmc-secret created
baremetalhost.metal3.io/openshift-worker-<num> created
```

Where `<num>` will be the worker number.

2. Power up and inspect the bare metal node.

```
[kni@provisioner ~]$ oc -n openshift-machine-api get bmh openshift-worker-<num>
```

Where `<num>` is the worker node number.

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
HARDWARE PROFILE	ONLINE	ERROR		
openshift-worker-<num>	OK	ready		ipmi://<out-of-
band-ip>	unknown	true		

3. Ensure the `PROVISIONING STATUS` is `ready` before provisioning the bare metal node.

```
[kni@provisioner ~]$ oc -n openshift-machine-api get bmh openshift-worker-<num>
```

Where `<num>` is the worker node number.

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
HARDWARE PROFILE	ONLINE	ERROR		
openshift-worker-<num>	OK	ready		ipmi://<out-of-
band-ip>	unknown	true		

4. Get a count of the number of worker nodes.

```
[kni@provisioner ~]$ oc get nodes
```

5. Get the machine set.

```
[kni@provisioner ~]$ oc get machinesets -n openshift-machine-api
```

NAME	DESIRED	CURRENT	READY	AVAILABLE	AGE
openshift-worker-0.example.com	1	1	1	1	55m
openshift-worker-1.example.com	1	1	1	1	55m
openshift-worker-2.example.com	1	1	1	1	55m

6. Increase the number of worker nodes by 1.

```
[kni@provisioner ~]$ oc scale --replicas=<num> machineset <machineset> -n  
openshift-machine-api
```

Replace **<num>** with the new number of worker nodes. Replace **<machineset>** with the name of the machine set from the previous step.

7. Check the status of the bare metal node.

```
[kni@provisioner ~]$ oc -n openshift-machine-api get bmh openshift-worker-<num>
```

Where **<num>** is the worker node number. The status changes from **ready** to **provisioning**.

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
HARDWARE PROFILE	ONLINE	ERROR		
openshift-worker-<num>-65tjz	OK	provisioning	openshift-worker-<num>-65tjz	
		unknown	ipmi://<out-of-band-ip>	true

The **provisioning** status remains until the OpenShift Container Platform cluster provisions the node. This may take 30 minutes or more. Once complete, the status will change to **provisioned**.

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
HARDWARE PROFILE	ONLINE	ERROR		
openshift-worker-<num>-65tjz	OK	provisioned	openshift-worker-<num>-65tjz	
		unknown	ipmi://<out-of-band-ip>	true

8. Once provisioned, ensure the bare metal node is ready.

```
[kni@provisioner ~]$ oc get nodes
```

NAME	STATUS	ROLES	AGE
provisioner.openshift.example.com v1.16.2	Ready	master	30h
openshift-master-1.openshift.example.com v1.16.2	Ready	master	30h
openshift-master-2.openshift.example.com v1.16.2	Ready	master	30h
openshift-worker-<num>.openshift.example.com v1.16.2	Ready	worker	3m27s

You can also check the kubelet.

```
[kni@provisioner ~]$ ssh openshift-worker-<num>
```

```
[kni@openshift-worker-<num>]$ journalctl -fu kubelet
```

Appending DNS records

Configuring Bind (Option 1)

Procedure

1. Login to the DNS server using `ssh`.
2. Suspend updates to all dynamic zones: `rndc freeze`.
3. Edit `/var/named/dynamic/example.com`.

```
$ORIGIN openshift.example.com.  
<OUTPUT_OMITTED>  
openshift-worker-1      A      <ip-of-worker-1>  
openshift-worker-2      A      <ip-of-worker-2>
```



Remove the provisioner as it is replaced by openshift-worker-2.

4. Increase the SERIAL value by 1.
5. Edit `/var/named/dynamic/1.0.10.in-addr.arpa`.



The filename `1.0.10.in-addr.arpa` is the reverse of the public CIDR example `10.0.1.0/24`.

6. Increase the SERIAL value by 1.

7. Enable updates to all dynamic zones and reload them: `rndc thaw`.

Configuring dnsmasq (Option 2)

Procedure

Append the following DNS record to the `/etc/hosts` file on the server hosting the `dnsmasq` service.

```
<OUTPUT_OMITTED>
<NIC2-IP> openshift-worker-1.openshift.example.com openshift-worker-1
<NIC2-IP> openshift-worker-2.openshift.example.com openshift-worker-2
```



Remove the `provisioner.openshift.example.com` entry as it is replaced by worker-2

Appending DHCP reservations

Configuring dhcpd (Option 1)

Procedure

1. Login to the DHCP server using `ssh`.
2. Edit `/etc/dhcp/dhcpd.hosts`.

```
host openshift-worker-2 {
    option host-name "worker-2";
    hardware ethernet <NIC2-mac-address>;
    option domain-search "openshift.example.com";
    fixed-address <ip-address-of-NIC2>;
}
```



Remove the provisioner as it is replaced by openshift-worker-2.

3. Restart the `dhcpd` service.

```
systemctl restart dhcpd
```

Configuring dnsmasq (Option 2)

Procedure

1. Append the following DHCP reservation to the `/etc/dnsmasq.d/example.dns` file on the server hosting the `dnsmasq` service.

```
<OUTPUT_OMITTED>
dhcp-host=<NIC2-mac-address>,openshift-worker-1.openshift.example.com,<ip-of-
worker-1>
dhcp-host=<NIC2-mac-address>,openshift-worker-2.openshift.example.com,<ip-of-
```



```
worker-2>
```



Remove the `provisioner.openshift.example.com` entry as it is replaced by `worker-2`

2. Restart the `dnsmasq` service.

```
systemctl restart dnsmasq
```

Deploying the provisioner node as a worker node using Metal3

After you have completed the prerequisites, perform the deployment process.

Procedure

1. Power off the node using `ipmitool` and confirm the provisioning node is powered off.

```
ssh <server-with-access-to-management-net>
# Use the user, password and Management net IP address to shutdown the system
ipmitool -I lanplus -U <user> -P <password> -H <management-server-ip> power off
# Confirm the server is powered down
ipmitool -I lanplus -U <user> -P <password> -H <management-server-ip> power status
Chassis Power is off
```

2. Get `base64` strings for the Out of band Management credentials. In this example, the user is `root` and the password is `calvin`.

```
# Use echo -ne, otherwise you will get your secrets with \n which will cause issues
# Get root username in base64
echo -ne "root" | base64
# Get root password in base64
echo -ne "calvin" | base64
```

3. Configure the BaremetalHost `bmh.yaml` file.

```
---
apiVersion: v1
kind: Secret
metadata:
  name: openshift-worker-2-bmc-secret
type: Opaque
data:
  username: ca2vdAo=
  password: MWAwTWdtdC0K
---
apiVersion: metal3.io/v1alpha1
```

```
kind: BareMetalHost
metadata:
  name: openshift-worker-2
spec:
  online: true
  bootMACAddress: <NIC1-mac-address>
  bmc:
    address: ipmi://<out-of-band-ip>
    credentialsName: openshift-worker-2-bmc-secret
```

4. Create the BaremetalHost.

```
./oc -n openshift-machine-api create -f bmh.yaml
secret/openshift-worker-2-bmc-secret created
baremetalhost.metal3.io/openshift-worker-2 created
```

5. Power up and inspect the node.

```
./oc -n openshift-machine-api get bmh openshift-worker-2
```

NAME	HARDWARE PROFILE	STATUS	ONLINE	PROVISIONING STATUS	ERROR	CONSUMER	BMC
openshift-worker-2		OK		inspecting			ipmi://<out-of-band-ip>

6. After finishing the inspection, the node is ready to be provisioned.

```
./oc -n openshift-machine-api get bmh openshift-worker-2
```

NAME	HARDWARE PROFILE	STATUS	ONLINE	PROVISIONING STATUS	ERROR	CONSUMER	BMC
openshift-worker-2	unknown	OK		ready			ipmi://<out-of-band-ip>

7. Scale the workers machineset. Previously, there were two replicas during original installation.

```
./oc get machineset -n openshift-machine-api
```

NAME	DESIRED	CURRENT	READY	AVAILABLE	AGE
openshift-worker-2	0	0			21h

```
./oc -n openshift-machine-api scale machineset openshift-worker-2 --replicas=3
```

8. The baremetal host moves to provisioning status. This can take as long as 30 minutes. You can follow the status from the node console.

```
oc -n openshift-machine-api get bmh openshift-worker-2
```

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
HARDWARE PROFILE	ONLINE	ERROR		
openshift-worker-2	OK	provisioning	openshift-worker-0-65tjz	
ipmi://<out-of-band-ip>		unknown	true	

9. When the node is provisioned it moves to provisioned status.

```
oc -n openshift-machine-api get bmh openshift-worker-2
```

NAME	STATUS	PROVISIONING STATUS	CONSUMER	BMC
HARDWARE PROFILE	ONLINE	ERROR		
openshift-worker-2	OK	provisioned	openshift-worker-2-65tjz	
ipmi://<out-of-band-ip>		unknown	true	

10. When the **kubelet** finishes initialization the node is ready for use. You can connect to the node and run **journalctl -fu kubelet** to check the process.

```
oc get node
```

NAME	STATUS	ROLES	AGE
VERSION			
openshift-master-0.openshift.example.com	Ready	master	30h
v1.16.2			
openshift-master-1.openshift.example.com	Ready	master	30h
v1.16.2			
openshift-master-2.openshift.example.com	Ready	master	30h
v1.16.2			
openshift-worker-0.openshift.example.com	Ready	worker	3m27s
v1.16.2			
openshift-worker-1.openshift.example.com	Ready	worker	3m27s
v1.16.2			
openshift-worker-2.openshift.example.com	Ready	worker	3m27s
v1.16.2			

Chapter 5. Appendix

In this section of the document, extra information is provided that is outside of the regular workflow.

5.1. Troubleshooting

Troubleshooting the installation is out of scope of the Deployment Guide. For more details on troubleshooting deployment, refer to our [Troubleshooting guide](#).

5.2. Creating DNS Records

Two options are documented for configuring DNS records:

- [On a DNS Server \(Bind\)](#)
- [Using dnsmasq](#)

5.2.1. Configuring Bind (Option 1)

Use Option 1 if access to the appropriate DNS server for the baremetal network is accessible or a request to your network admin to create the DNS records is an option. If this is not an option, skip this section and go to section Create DNS records using dnsmasq (Option 2).

Create a subzone with the name of the cluster that is going to be used on your domain. In our example, the domain used is **example.com** and the cluster name used is **openshift**. Make sure to change these according to your environment specifics.

Procedure

1. Login to the DNS server using **ssh**.
2. Suspend updates to all dynamic zones: **rndc freeze**.
3. Edit **/var/named/dynamic/example.com**.

```
$ORIGIN openshift.example.com.
$TTL 300          ; 5 minutes
@ IN SOA dns1.example.com. hostmaster.example.com. (
    2001062501    ; serial
    21600         ; refresh after 6 hours
    3600          ; retry after 1 hour
    604800        ; expire after 1 week
    86400 )       ; minimum TTL of 1 day
;
api                A        <api-ip>
ns1                A        <dns-vip-ip>
$ORIGIN apps.openshift.example.com.
*                  A        <wildcard-ingress-lb-ip>
$ORIGIN openshift.example.com.
```

provisioner	A	<NIC2-ip-of-provision>
openshift-master-0	A	<NIC2-ip-of-openshift-master-0>
openshift-master-1	A	<NIC2-ip-of-openshift-master-1>
openshift-master-2	A	<NIC2-ip-of-openshift-master-2>
openshift-worker-0	A	<NIC2-ip-of-openshift-worker-0>
openshift-worker-1	A	<NIC2-ip-of-openshift-worker-1>

4. Increase the `serial` value by 1.
5. Edit `/var/named/dynamic/1.0.10.in-addr.arpa`.

```
$ORIGIN 1.0.10.in-addr.arpa.
$TTL 300
@ IN SOA dns1.example.com. hostmaster.example.com. (
    2001062501 ; serial
    21600      ; refresh after 6 hours
    3600       ; retry after 1 hour
    604800     ; expire after 1 week
    86400 )    ; minimum TTL of 1 day
;
126 IN PTR provisioner.openshift.example.com.
127 IN PTR openshift-master-0.openshift.example.com.
128 IN PTR openshift-master-1.openshift.example.com.
129 IN PTR openshift-master-2.openshift.example.com.
130 IN PTR openshift-worker-0.openshift.example.com.
131 IN PTR openshift-worker-1.openshift.example.com.
132 IN PTR api.openshift.example.com.
133 IN PTR ns1.openshift.example.com.
```



In this example, the IP addresses 10.0.1.126-133 are pointed to the corresponding fully qualified domain name.



The filename `1.0.10.in-addr.arpa` is the reverse of the public CIDR example `10.0.1.0/24`.

6. Increase the `serial` value by 1.
7. Enable updates to all dynamic zones and reload them: `rndc thaw`.

5.2.2. Configuring dnsmasq (Option 2)

To create DNS records, open the `/etc/hosts` file and add the NIC2 (baremetal net) IP followed by the hostname. In our example, the domain used is `example.com` and the cluster name used is `openshift`. Make sure to change these according to your environment specifics.

Procedure

1. Edit `/etc/hosts` and add the NIC2 (baremetal net) IP followed by the hostname.

```
cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
<NIC2-IP> provisioner.openshift.example.com provisioner
<NIC2-IP> openshift-master-0.openshift.example.com openshift-master-0
<NIC2-IP> openshift-master-1.openshift.example.com openshift-master-1
<NIC2-IP> openshift-master-2.openshift.example.com openshift-master-2
<NIC2-IP> openshift-worker-0.openshift.example.com openshift-worker-0
<NIC2-IP> openshift-worker-1.openshift.example.com openshift-worker-1
<API-IP>  api.openshift.example.com api
<DNS-VIP-IP> ns1.openshift.example.com ns1
```

2. Open the appropriate `firewalld` DNS service and reload the rules.

```
systemctl restart firewalld
firewall-cmd --add-service=dns --permanent
firewall-cmd --reload
```

5.3. Creating DHCP reservations

Two options are documented for configuring DHCP:

- [On dhcpd \(Option 1\)](#)
- [Using dnsmasq \(Option 2\)](#)

5.3.1. Configuring dhcpd (Option 1)

Use Option 1 if access to the appropriate DHCP server for the baremetal network is accessible or a request to your network admin to create the DHCP reservations is an option. If this is not an option, skip this section and go to section Create DHCP records using dnsmasq (Option 2).

1. Login to the DHCP server using `ssh`.
2. Edit `/etc/dhcp/dhcpd.hosts`.

```
host provisioner {
    option host-name "provisioner";
    hardware ethernet <mac-address-of-NIC2>;
    option domain-search "openshift.example.com";
    fixed-address <ip-address-of-NIC2>;
}
host openshift-master-0 {
    option host-name "openshift-master-0";
    hardware ethernet <mac-address-of-NIC2>;
    option domain-search "openshift.example.com";
    fixed-address <ip-address-of-NIC2>;
}
```

```

host openshift-master-1 {
    option host-name "openshift-master-1";
    hardware ethernet <mac-address-of-NIC2>;
    option domain-search "openshift.example.com";
    fixed-address <ip-address-of-NIC2>;
}

host openshift-master-2 {
    option host-name "openshift-master-2";
    hardware ethernet <mac-address-of-NIC2>;
    option domain-search "openshift.example.com";
    fixed-address <ip-address-of-NIC2>;
}

host openshift-worker-0 {
    option host-name "openshift-worker-0";
    hardware ethernet <mac-address-of-NIC2>;
    option domain-search "openshift.example.com";
    fixed-address <ip-address-of-NIC2>;
}

host openshift-worker-1 {
    option host-name "openshift-worker-1";
    hardware ethernet <mac-address-of-NIC2>;
    option domain-search "openshift.example.com";
    fixed-address <ip-address-of-NIC2>;
}

```

3. Restart the `dhcpcd` service.

```
systemctl restart dhcpcd
```

5.3.2. Configuring dnsmasq (Option 2)

Set up `dnsmasq` on a server that can access the baremetal network.

Procedure

1. Install `dnsmasq`.

```
dnf install -y dnsmasq
```

2. Change to the `/etc/dnsmasq.d` directory.

```
cd /etc/dnsmasq.d
```

3. Create a file that reflects your OpenShift cluster appended by `.dns`.

```
touch <filename>.dns
```

4. Open the appropriate `firewalld` DHCP service.

```
systemctl restart firewalld  
firewall-cmd --add-service=dhcp --permanent  
firewall-cmd --reload
```

5. Define DNS configuration file

IPv4

Here is an example of the `.dns` file for IPv4.

```
domain-needed  
bind-dynamic  
bogus-priv  
domain=openshift.example.com  
dhcp-range=<baremetal-net-starting-ip,baremetal-net-ending-ip>  
#dhcp-range=10.0.1.4,10.0.14  
dhcp-option=3,<baremetal-net-gateway-ip>  
#dhcp-option=3,10.0.1.254  
resolv-file=/etc/resolv.conf.upstream  
interface=<nic-with-access-to-baremetal-net>  
#interface=em2  
server=<ip-of-existing-server-on-baremetal-net>  
  
#Wildcard for apps -- make changes to cluster-name (openshift) and domain  
(example.com)  
address=/.apps.openshift.example.com/<wildcard-ingress-lb-ip>  
  
#Static IPs for Masters  
dhcp-host=<NIC2-mac-address>,provisioner.openshift.example.com,<ip-of-provisioner>  
dhcp-host=<NIC2-mac-address>,openshift-master-0.openshift.example.com,<ip-of-  
openshift-master-0>  
dhcp-host=<NIC2-mac-address>,openshift-master-1.openshift.example.com,<ip-of-  
openshift-master-1>  
dhcp-host=<NIC2-mac-address>,openshift-master-2.openshift.example.com,<ip-of-  
openshift-master-2>  
dhcp-host=<NIC2-mac-address>,openshift-worker-0.openshift.example.com,<ip-of-  
openshift-worker-0>  
dhcp-host=<NIC2-mac-address>,openshift-worker-1.openshift.example.com,<ip-of-  
openshift-worker-1>
```


IPv6

Here is an example of the `.dns` file for IPv6.

```
strict-order
bind-dynamic
bogus-priv
dhcp-authoritative
dhcp-range=baremetal,<baremetal-IPv6-dhcp-range-start>,<baremetal-IPv6-dhcp-range-
end>,<range-prefix>
dhcp-option=baremetal,option6:dns-server,[<IPv6-DNS-Server>]

resolv-file=/etc/resolv.conf.upstream
except-interface=lo
dhcp-lease-max=81
log-dhcp

domain=openshift.example.com,<baremetal-IPv6-cidr>,local

# static host-records
address=/.apps.openshift.example.com/<wildcard-ingress-lb-ip>
host-record=api.openshift.example.com,<api-ip>
host-record=ns1.openshift.example.com,<dns-ip>
host-record=openshift-master-0.openshift.example.com,<ip-of-openshift-master-0>
host-record=openshift-master-1.openshift.example.com,<ip-of-openshift-master-1>
host-record=openshift-master-2.openshift.example.com,<ip-of-openshift-master-1>
# Registry
host-record=registry.openshift.example.com,<ip-of-registry-server>

#Static IPs for Masters
dhcp-host=<baremetal-nic-duid>,openshift-master-0.openshift.example.com,<ip-of-
openshift-master-0>
dhcp-host=<baremetal-nic-duid>,openshift-master-1.openshift.example.com,<ip-of-
openshift-master-1>
dhcp-host=<baremetal-nic-duid>,openshift-master-2.openshift.example.com,<ip-of-
openshift-master-2>
```

6. Create the `resolv.conf.upstream` file to provide DNS forwarding to an existing DNS server for resolution to the outside world.

```
search <domain.com>
nameserver <ip-of-my-existing-dns-nameserver>
```

7. Restart the `dnsmasq` service.

```
systemctl restart dnsmasq
```

8. Verify the `dnsmasq` service is running.

```
systemctl status dnsmasq
```